DATE: 25TH OCTOBER 2019

TECHNICAL ESSAY

TOPIC: THERMAL PROTECTION SYSTEM OF SPACECRAFT

NAME: DEBJYOTI SENGUPTA

PROGRAM: MEng MECHANICAL ENGINEERING

YEAR OF INTAKE: 2019 – 20

Review Paper

Thermal Protection System of Spacecrafts – A Necessity for Deep Space Exploration

Debjyoti Sengupta [a]

[a] Undergraduate Student, Mechanical Engineering, School of Engineering, University of Southampton, UK

ARTICLE INFO:

Keywords – *Thermal Protection Systems (TPSs), Spacecraft, Ablative, Composites, Atmospheric Re-entry, Thermal Analysis*

TABLE OF CONTENTS:

i. Abstract **2**
ii. Introduction **3**

1. Categorisation of TPS ....
1.1 Ablative Systems ....
1.2 Radiative Systems ....
1.3 Heat Sink System ....
1.4 Hot Structure ....
1.5 Transpiration Cooling ....
2. Thermal Analysis ....
2.1 Conductive Heat Exchange ....
2.2 Radiative Heat Exchange ....
2.2.1 Radiative View Factors ....
2.2.2 Effective Emittance ....
3. Research Advances in TPS ....
4. Problems to be Faced and Possible Long-Term Solutions ....
5. Way Forward and Inference ....
iii. References ....
iv. Appendix A ....
v. Appendix B ....
vi. Appendix C ....
vii. Appendix D ....

ABSTRACT:

The aim of the review article is to study the thermal protection system, more specifically – the passive thermal protection system used for the launch vehicles and re-entry vehicles. The review article throws light on recent advances, problems faced and the way forward in the research on the thermal protection system of spacecrafts as the thermal protection system is an important component in spacecraft systems engineering, especially during manned missions to low earth orbit (LEO), the Moon and beyond.

INTRODUCTION:

When a spacecraft must land on a planet's surface at moderate to hypersonic velocity, the excess of kinetic energy which the spacecraft possesses when it is in orbit has to be dissipated. The use of fuel to power atmospheric re-entry is prohibitive with keeping mass of the spacecraft in mind. Thus, the re-entry of the spacecraft must be facilitated by aerodynamic drag. The aerodynamic drag slows down the vehicle but at the same time, due to friction with the ions present in the atmosphere and the highly changing aerodynamic environment, generates huge amount of heat on the vehicle which may cause the vehicle to break-up during atmospheric re-entry of the spacecraft; as was evident from the Space Shuttle Columbia Tragedy, where the Space Shuttle Columbia broke up over the state of Texas during atmospheric re-entry because of damaged sustained by the thermal protection system on the leading edge of the spacecraft's left wing, thus killing all the seven crew members of the NASA STS-107 mission on February 1, 2003.

Initially, the atmosphere's density is very low and free molecular flow across the spacecraft. Then, as the vehicle progresses further into the planet's atmosphere, a transitional flow is formed with thick viscous shock waves around the spacecraft. Finally, once the spacecraft is sufficiently low in the atmosphere, a continuum flow region with velocities still hypersonic is encountered, which causes very large heat fluxes on the surface of the spacecraft over the period of atmospheric re-entry.

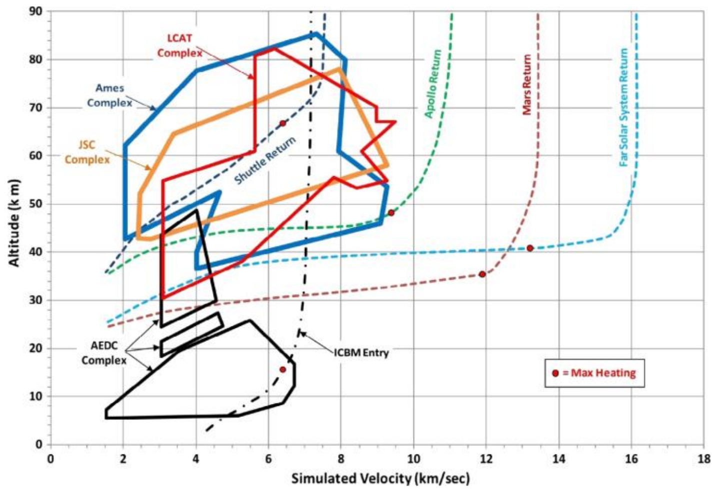


Figure 1: Comparison of NASA, AEDC, and Boeing arc jet facility performance with several nominal Earth return trajectories. The testing capabilities – based on current arc-jet facilities – for TPS materials are also evidenced. The Ames Research Center (ARC), the Johnson Space Center (JSC), and the Boeing Large Core Arc Tunnel (LCAT) simulation envelopes are at the higher altitudes, which capture the major portion of the Space Shuttle and other manned vehicle trajectories. The Arnold Engineering Development Center (AEDC) facilities are primarily in the lower altitude regime, suitable for weapon trajectory simulation, and capture the maximum heating point of an ICBM entry.

The Thermal Protection System (TPS) of a space vehicle is the hardware used to guarantee the structural integrity of the spacecraft and maintain proper internal temperatures during the re-entry phase of the mission. The peak heat flux determines the selection of the type of thermal protection system to be used and the thermal protection material which will help the spacecraft maintain its structural integrity during the re-entry phase. [1]

Re-entry into the Earth's atmosphere has been studied in detail since the development of ballistic missiles and the Apollo program. It is still a subject of major research as well as a matter of major concern in upcoming space vehicles like the NASA Orion, Boeing CST – 100 Starliner, SpaceX Dragon and others.

Figure 2: Prototype (5 m diameter) of heat shield for Orion Crew Exploration Vehicle (CEV) (NASA)

1. CATEGORISATION OF TPS:

The TPS can be categorized into three major categories – passive, semi-passive and active. Passive TPS comprises of Ablative Systems and Radiative Systems. Semi-passive TPS are made up of two major systems – Heat Sink TPS and Hot Structures. Transpiration cooling is the only known form of Active TPS.

1.1 Ablative Systems

Ablative TPS have been used effectively in most non-reusable entry vehicles due to their high efficiency. These sacrificial materials dissipate the incident thermal energy through change of phase (usually form plasma streaks) and loss of mass of the entry vehicle. The great amount of energy absorbed during the phase change by the ablative material during atmospheric entry/re-entry reduces the heat flux experienced by the structure of the spacecraft. An ablative TPS is based on sublimation, melting, or pyrolysis of the heat shield and the removal of the material/products by the outer stream. [2]

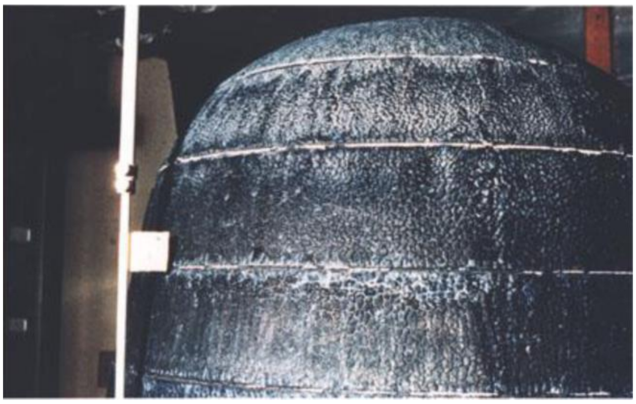


Figure 3: Photograph of a charred ablative heat shield of re-entry vehicle (NASA)

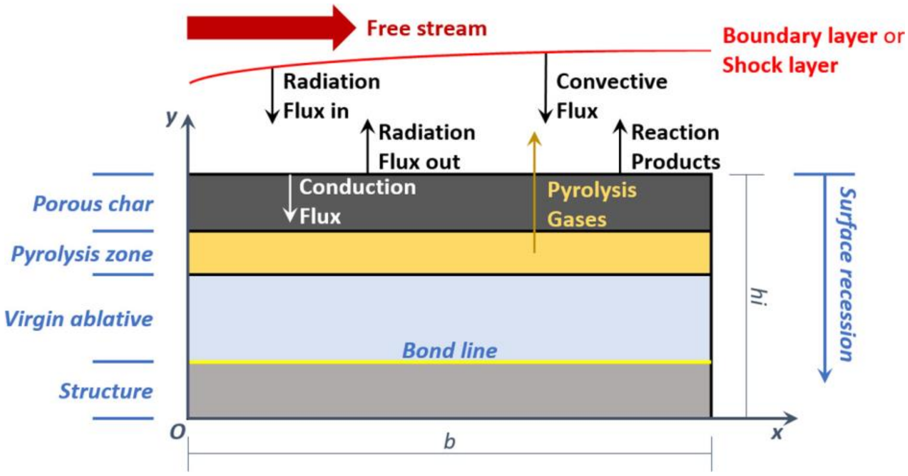


Figure 4: Structure Scheme and Heat Fluxes for Charring Ablative Heat Shield

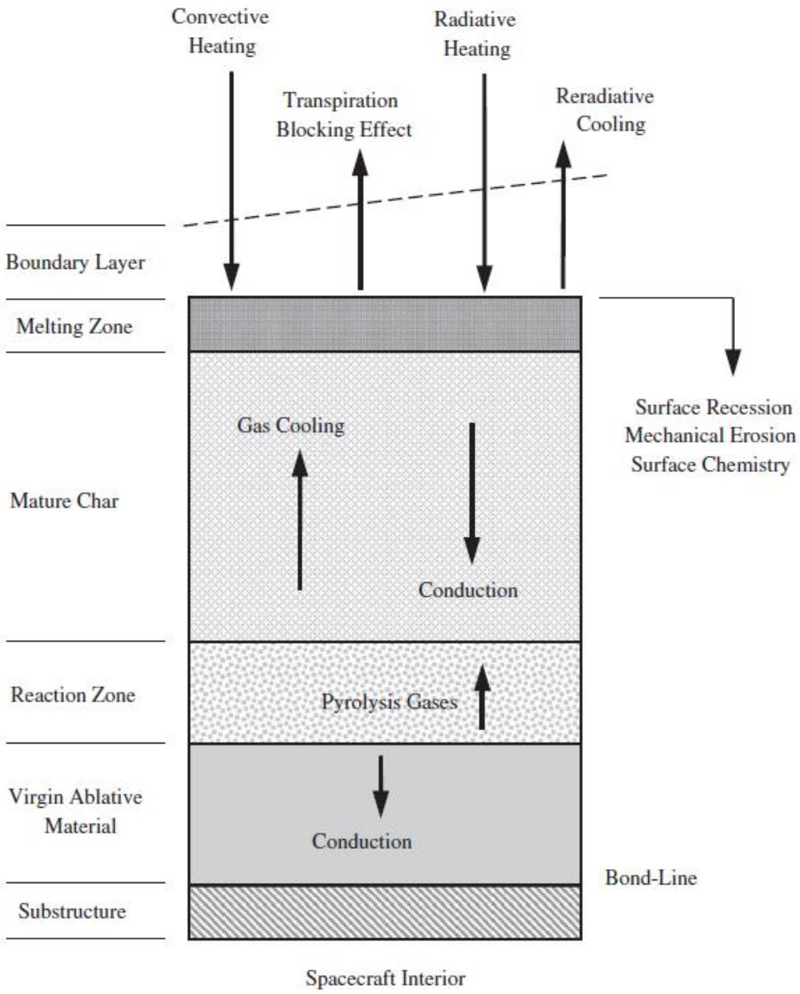


Figure 5: Schematic of the charring ablation phenomena

1.2 Radiative Systems

Radiative TPS re-emit the energy received from the received from the environment in the form of thermal radiation by virtue of high surface temperature of the system/material. The heat flux can be described by the expression,

$$q = \varepsilon\sigma T^4 \dots\dots\dots\dots (1)$$

where ε is the emissivity of the surface and σ is the Stefan-Boltzmann Constant. This means that the upper temperature limit of the material can determine the maximum heat energy radiated by the material. The application of the radiative TPS are then limited to mild entry environment as the ones encountered by the NASA STS, commonly known as the Space Shuttle. Most of the convective and radiative energy received from the hot boundary layer gases are re-radiated but part of the absorbed energy can be conducted into the vehicle and its structure. To avoid this, we require a high emissivity and good thermal insulation material for the same. [3]

Figure 6: Space Shuttle Thermal Protection Tile – an example of radiative TPS

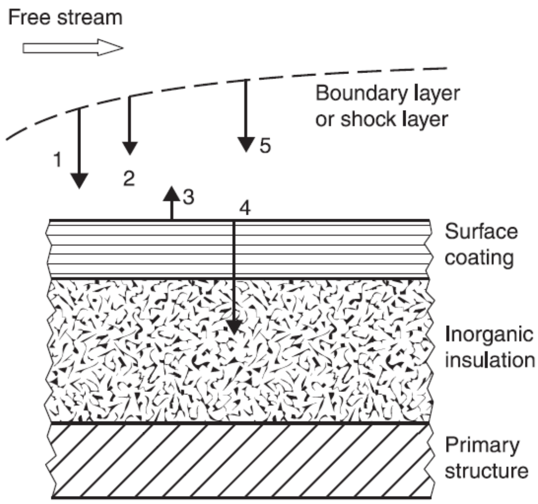


Figure 7: Energy Interactions on a reusable TPS Material. We must note that radiative TPS are the most common type of reusable TPS used in spacecrafts.

1.3 Heat Sink System

A heat sink is the simplest form of an absorptive thermal protection system. Heat sink systems consists of a relatively thick layer of a material with high thermal capacity to absorb a heat input. When the surface is heated, a part of the energy is radiated away, and the remaining energy is absorbed by the structure of the spacecraft. If a heat sink structure is heated over a long period of time, the structure can get overheated. Thus, heat sink TPSs are used with moderate heat flux for transient situation.

1.4 Hot Structures

Hot Structures type of TPS have been developed for unmanned experimental spacecraft for orbital spaceflight missions intended to demonstrate technologies that will render the spacecraft as reusable, which is becoming a necessary requirement for any spacecraft with every passing day. The hot structure is lighter in weight and require less maintenance than those materials joined to a hot or cold substructure. However, the development of hot structures requires a thorough understanding of the material's thermal and mechanical performance in an extreme environment, operating at temperatures of 1500℃ to 3000℃, depending on the application and the mission scenario. [3]

1.5 Transpiration Cooling

In transpiration cooling systems, a fluid is injected through a porous medium into the boundary layer. The structure is cooled down by means of two main mechanisms. First, heat is conducted to the coolant as it flows through the structure and second, when the coolant is ejected it cools and thickens the boundary layer. Water, carbon dioxide, ammonia are some of the most acceptable coolants. This form of system is mostly used in the propulsion engines of the launch vehicle (LV), where transpiration cooling keeps the engine in the most appropriate temperature for the start-up of the main engines of the LV during T-5 seconds to T-0 seconds in the countdown to the lift-off.

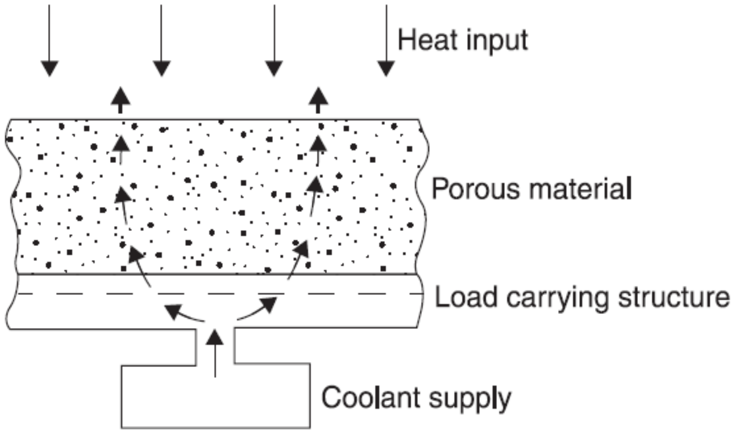


Figure 8: A typical transpiration cooling system

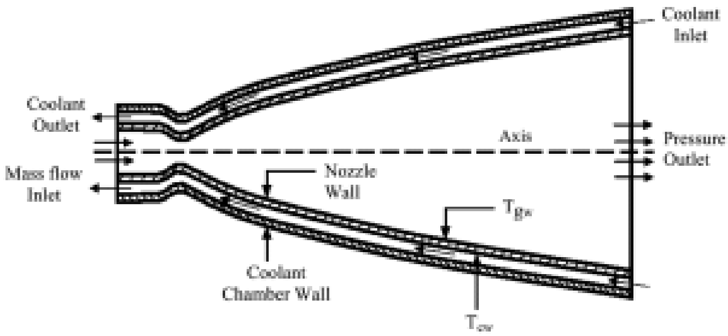


Figure 9: Application of transpiration cooling for a rocket engine like the Aerojet Rocketdyne RS-25 (also commonly known as Space Shuttle Main Engine (SSME))

2. THERMAL ANALYSIS – A TOOL TO DETERMINE THERMAL DESIGN:

Spacecrafts are very complex structures within which temperatures are a varying function of location and time. This function of location and time is a continuous function, which means temperature varies continuously. Calculating these temperature fields in rigorous detail is practically impossible. In order to make progress, we need to simplify this engineering problem. This is possible by generating an approximate representation of the spacecraft that is amenable for mathematical treatment. This is called as the Thermal Mathematical Model (TMM).

In order to create a TMM, the spacecraft is assumed to be composed of various discrete regions within which temperature gradients can be neglected. These regions are called as isothermal nodes. Each node is characterized by a temperature, heat dissipation (if any), radiative and conductive surfaces with surrounding nodes. Nodes that are exposed to the space directly will compose of radiative interface with the external environment.

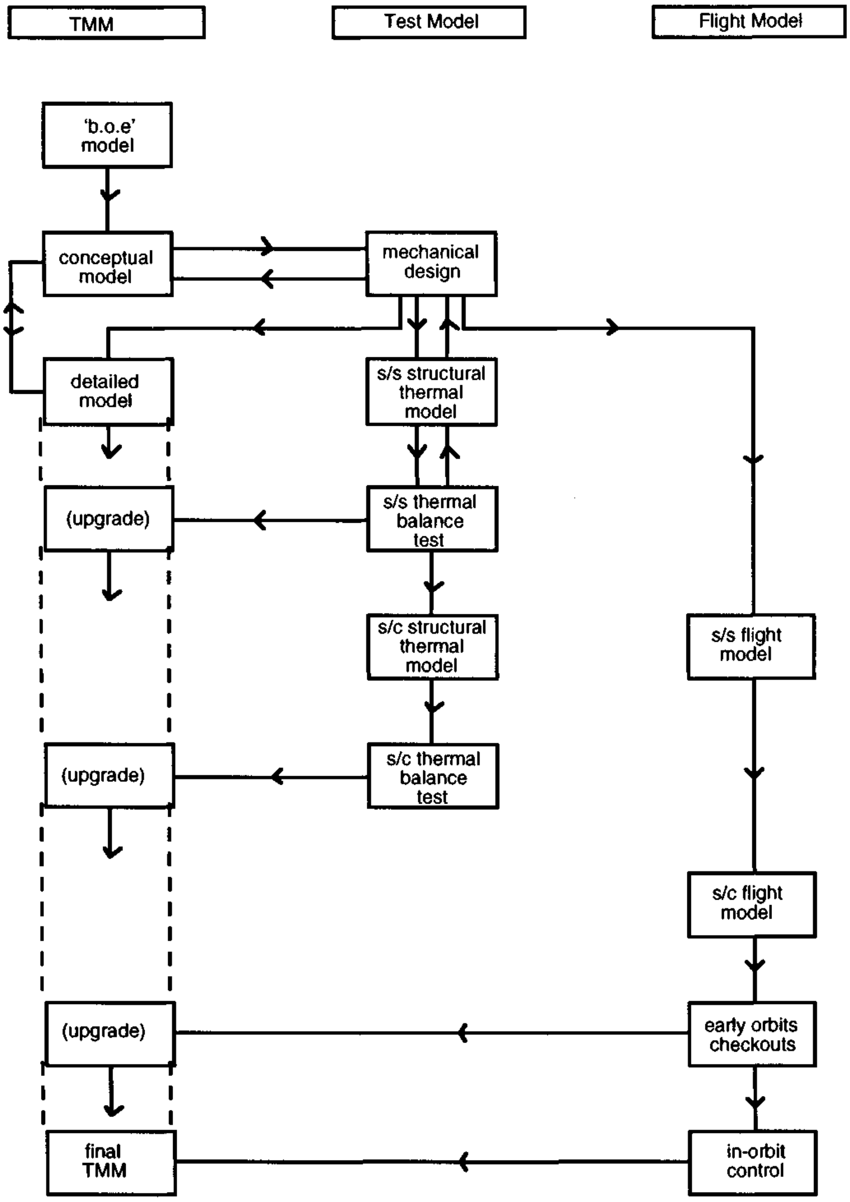


Figure 10: A strategic flow chart to determine the thermal design for a spacecraft [4]. The flow chart is discussed in detail in Appendix D.

2.1 Conductive Heat Exchange

The conductive heat flow Q_c, is given by,

$$Q_c = \frac{\lambda A}{l} \Delta T \ldots\ldots\ldots\ldots (2)$$

where λ is the thermal conductivity, A is the cross-sectional area, l is the conductive path length and ΔT is the temperature difference. The term $\frac{\lambda A}{l}$ is known as the thermal conductance, h_c , and the temperature difference can be written as,

$$\Delta T = \frac{Q_c}{h_c} \ldots\ldots\ldots\ldots (3)$$

In most engineering applications, A and possibly λ may vary significantly along the path length. If the conductive path is considered as several discrete conductive paths connected in series, the temperature difference can be rewritten as,

$$\Delta T = Q_c \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3} + \cdots \right) = \frac{Q_c}{h_c} \ldots\ldots\ldots\ldots (4)$$

and hence the effective thermal conductance, h_c , for the path can be found from

$$\frac{1}{h_c} = \left(\frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3} + \cdots \right) \ldots\ldots\ldots\ldots (5)$$

For a spacecraft composed of n isothermal nodes, the heat conducted from the ith node to jth node is given by,

$$Q_{c_{ij}} = h_{ij}\left(T_i - T_j\right) \ldots\ldots\ldots\ldots (6)$$

where h_{ij} is the effective conductance between nodes i and j and T_i and T_j are the temperatures of the ith and jth nodes, respectively. [4] [5]

2.2 Radiative Heat Exchange

Radiative heat exchange between two surfaces is determined by three important parameters – the surface temperature of the nodes, the radiative view factors and the properties of the surface. For diffuse surfaces, the amount of radiation leaving a surface/node i and absorbed by a surface/node j can be shown to be of the form,

$$Q_{r_{ij}} = A_i F_{ij} \varepsilon_{ij} \sigma\left(T_i^4 - T_j^4\right) \ldots\ldots\ldots\ldots (7)$$

where A_i is the area of the node i, F_{ij} is the radiative view factor of node j as seen from node i and ε_{ij} is a parameter known as the effective emittance. Note that it is assumed in the above that the value of the radiative view factor F_{ij} remains constant over the node i.

2.2.1 Radiative View Factors:

The radiative view factor F_{ij} is defined as the fraction of the radiation leaving one surface that is intercepted by another. From a node i inside a spacecraft, the sum of the radiative view factors to the surrounding nodes must be unity, given by the expression,

$$\sum_{j=1}^{k} F_{ij} = 1 \ldots\ldots\ldots\ldots (8)$$

where k is equal to the number of surrounding surfaces.

Consider two surfaces A_1 and A_2 as indicated in the figure below. Let us assume that these surfaces are diffuse surface. The radiation emitted from A_1 to A_2, per unit solid angle and per unit area of A_1, can be given by the equation,

$$I_1 = I_0 \cos\phi_1 \ldots\ldots\ldots\ldots (9)$$

where I_0 is the radiation intensity normal to A_1 and ϕ_1 is the solid angle.
For the elementary surfaces δA_1 and δA_2, the infinitesimally small amount of radiation leaving δA_1 and falling on δA_2 is given by,

$$\delta Q_{r_{12}} = I_0 \delta A_1 \cos\phi_1 \frac{\delta A_2 \cos\phi_2}{s^2} \dots\dots\dots\dots (10)$$

Hence, the total radiation reaching A_2 from A_1 is given by,

$$Q_{r_{12}} = I_0 \int_{A_1} \int_{A_2} \frac{\cos\phi_1 \cos\phi_2}{s^2}\, dA_1\, dA_2 \dots\dots\dots\dots (11)$$

However, from Equation (9), we can easily show the total radiation leaving A_1 by the following equation:

$$Q_{r_{tot}} = A_1 I_0 \int_0^{\frac{\pi}{2}} 2\pi \sin\phi \cos\phi \; d\phi = A_1 \pi I_0 \dots\dots\dots\dots (12)$$

The view factor in a general form can be expressed by the equation below:

$$A_i F_{ij} = \int_{A_i} \int_{A_j} \frac{\cos\phi_1 \cos\phi_2}{\pi s^2}\, dA_i\, dA_j \dots\dots\dots\dots (13)$$

From an examination of the symmetry of the equation, we can deduce an important reciprocity relationship, which can be shown as

$$A_i F_{ij} = A_j F_{ji} \dots\dots\dots\dots (14)$$

2.2.2 Effective Emittance

The effective emittance between two nodes has a complicated dependence on surface optical properties, mutual reflections and reflections through the nearby surfaces. In the simple case in which all surfaces are perfectly black, the effective emittance reduces to the trivial result $\varepsilon_{ij} = 1$. Specular surfaces and/or emittance values less than unity give rise to more complicated expressions for ε_{ij}. For the simplest case of two parallel, diffuse surfaces, separated by a distance that is negligible in comparison to the surface area of the nodes, we can obtain the following expression:

$$\varepsilon_{ij} = \frac{\varepsilon_i \varepsilon_j}{\varepsilon_i + \varepsilon_j - \varepsilon_i \varepsilon_j} \dots\dots\dots\dots (15)$$

Although this is a special case, seldom encountered in practice, errors introduced by it are extremely small, provided the surfaces are diffuse in nature and have relatively high individual effective emittance ε values. This expression is commonly used to calculate practical effective emittance values and is acceptable when the spacecraft interior structure is painted.
Let us consider, at a particular time, the heat balance of the ith node of a TMM consisting of n nodes. The total (net) heat absorbed $Q_{abs,i}$ by the surface i per unit time is given by:

$$Q_{ext,i} + Q_i - \sigma \varepsilon_i A_{space,i} T_i^4 - \sum_{j=1}^{n} h_{ij}(T_i - T_j) - \sigma \sum_{j=1}^{n} A_i F_{ij} \varepsilon_{ij} \left(T_i^4 - T_j^4\right) \dots\dots\dots\dots (16)$$

where

$$Q_{ext,i} = J_s \alpha_i A_{solar,i} + J_p \alpha_i A_{planetary,i} + J_a \alpha_i A_{albedo,i} \dots\dots\dots\dots (17)$$

is the external heat input. $A_{solar,i}$, $A_{planetary,i}$, $A_{albedo,i}$ are the effective areas receiving direct solar, albedo and planetary radiation respectively. $A_{space,i}$ is the effective area with an unobstructed view of space and Q_i is the internal heat dissipation. If the mass and specific heat of the node can be represented as m_i and C_i , respectively, the heat balance equation can be written as:

$$m_i C_i \frac{dT_i}{dt} = Q_{ext,i} + Q_i - \sigma \varepsilon_i A_{space,i} T_i^4 - \sum_{j=1}^{n} h_{ij}(T_i - T_j) - \sigma \sum_{j=1}^{n} A_i F_{ij} \varepsilon_{ij} \left(T_i^4 - T_j^4\right) \dots \dots \dots \dots (18)$$

The thermal behaviour of the complete spacecraft is thus described by the set of n simultaneous non-linear differential equations as above, with i varying from 1 to n. It should be noted that, in real-life scenario, $Q_{ext,i}$ and Q_i will be varying functions dependent on time. We can observe that the above set of equations is not amenable to analytical solution, which means we will require numerical methods to solve the equations above. The most common approach to solve the given set of equations is to linearize them in the following way. [5] [6]

If $T_{i,0}$ is the temperature of the ith node at some arbitrary time, t_0, the temperature of that node a short while, δt , later is given by

$$T_i = T_{i,0} + \delta T_{i,0} \quad where \quad \delta T_{i,0} = \frac{dT_{i,0}}{dt} \delta t \dots \dots \dots \dots (19)$$

From the first of these equations and assuming that $\delta T_{i,0}$ is small compared with $T_{i,0}$ we see that

$$T_i^4 = \left(T_{i,0} + \delta T_{i,0}\right)^4 \approx T_{i,0}^4 + 4T_{i,0}^3 \delta T_{i,0} \dots \dots \dots \dots (20)$$

and hence that

$$T_i^4 \approx T_i\left(4T_{i,0}^3\right) - 3T_{i,0}^4 \dots \dots \dots \dots (21)$$

In the special case of steady-state calculations, where $Q_{ext,i}$ and Q_i are constant and $\frac{dT}{dt}$ is zero, which will linearize the equation (21) and substitute in (18). A similar approach can be used to make non-steady/transient calculations.

3. RESEARCH ADVANCES IN TPS:

In the research article [7], the authors have made a genuine attempt to estimate heat fluxes on the surface of advanced materials with known thermal and thermo-kinetic properties using the approach of inverse methods. Through indirect measurements like these, we are able to formulate solutions to inverse heat transfer problems. By solving such inverse problems, the boundary conditions and the unsteady temperature distribution is reconstructed using interior temperature measurements in structures as shown by the data provided by the sensors. This data is discussed in Appendix C using equations.

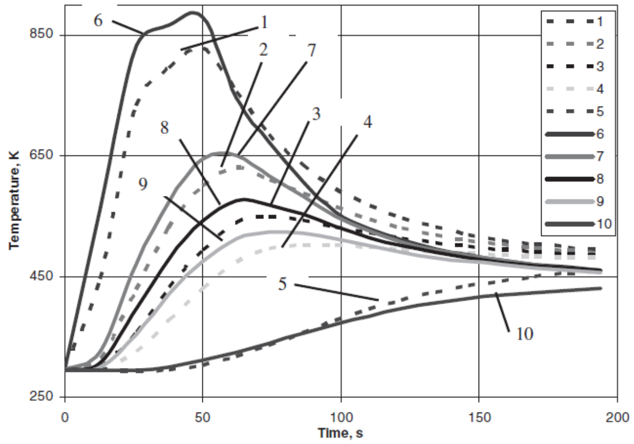


Figure 11: Comparing calculated and measured temperatures

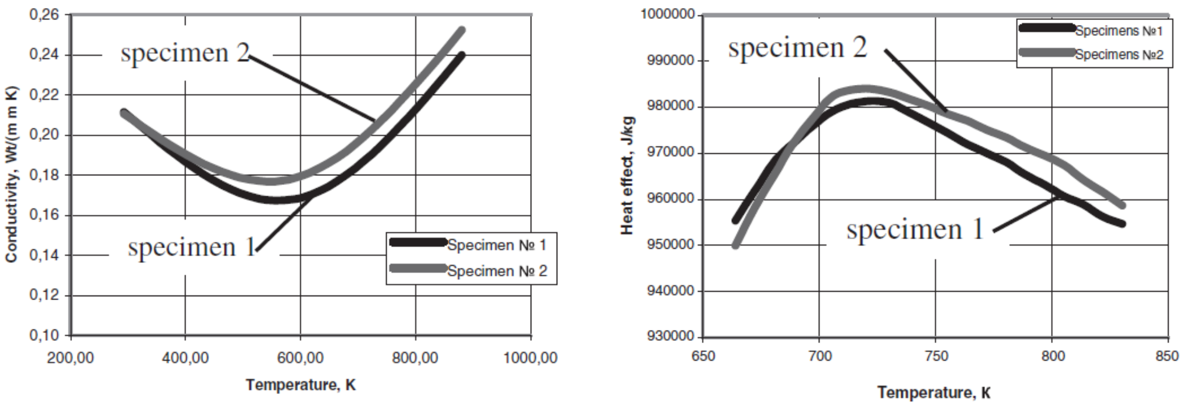


Figure 12 and 13: Estimated values of thermal conductivity and heat effect of TPS

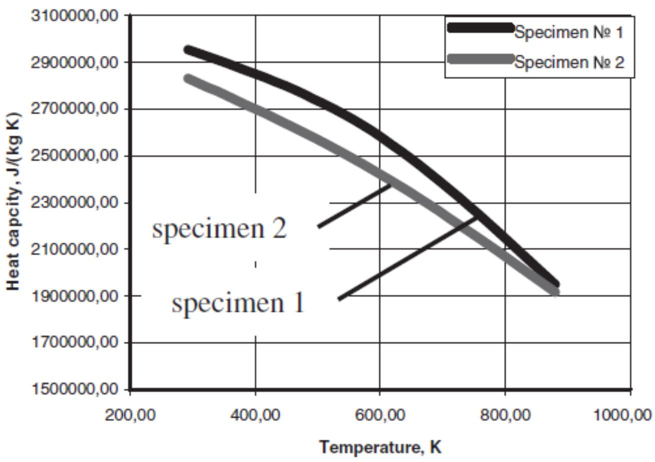


Figure 14: Estimated values of heat capacity

One of the most commonly used TPS is the ablative TPS. In fact, till date, ablative TPS remain the primary thermal protection used for various aerospace applications. Various researchers across the world are trying to find out new ablative materials which can be used as an efficient TPS. Researchers from Italy [8] have come up with a probabilistic sizing tool for entry vehicle ablative TPS. An implicit ablation and thermal response software was developed to analyse charring ablative TPS for entry vehicles. A statistical monitor embedded into the tool, using the Monte Carlo analysis, allows a simulation to run over stochastic series, thus helping to analyse and decide the right TPS for a given mission. The thermal model obtained using the Monte Carlo analysis is discussed in detail in Appendix C.

The research article [9] is a report which comprises the results of theoretical and experimental research aimed at development of special devices, to be precise – Inflatable Braking Devices (IBD), to increase the effectiveness of the TPS of the spacecraft during descent or atmospheric re-entry. Through ballistic analysis, the report describes the dependency of ballistic coefficients of descent vehicle (DV) vs. payload masses regarding producible modern materials. The report also provides a comparison between possible mass properties of the IBD-based DV and analogous ballistic properties of a real DV carrying out ballistic descent.

Researchers at NASA Langley Research Centre and at India have carried out numerical analysis and tried to study transient response of advanced TPS for atmospheric re-entry. In the research article [10], heat diffusion through a planar ablative TPS is investigated by numerical analysis by modelling the problem as one-dimensional transient heat conduction problem in Cartesian co-ordinates subject to the adiabatic back wall and aerodynamic heating on the other surface. Ablation is modelled as a Stefan-type wherein layers of material are immediately removed upon melt after reaching ablative temperature. By developing a FORTRAN code is developed to solve the set of equations using Tri-Dimensional Matrix Algorithm (TDMA), we obtain co-relations which predict the parameters quite well and errors are within acceptable range. As per the NASA technical report D-1370 [11] , a computer program; which is capable of providing a numerical analysis of charring ablator, impregnated ceramic, subliming, or heat-sink thermal protection systems, subjected to a severe heating environment; is used to solve equations for the transfer of heat through thermal protection shields are derived in finite difference form. Some usual results are presented in the technical note. Limited comparisons with experimental results are made. For materials whose thermal properties are well-known, good agreement has been obtained between calculated and experimental results.

4. PROBLEMS TO BE FACED AND POSSIBLE LONG-TERM SOLUTIONS:

In the modern era, to determine the heat protection design of the spacecraft, there is a common tendency to conduct comprehensive analysis and estimation of the thermal environment, heat protection and the thermal structure using computer system.
The heat protection design for spacecraft is a complicated process dealing with many scientific fields, such as materials, solid mechanics, aerothermodynamics and others. Some of these factors are uncertain, especially the consistence conditions between two fields are usually difficult to know. In order to ensure the safety of the re-entry vehicle, conservative designs are usually selected. In [12], the authors have presented a new format of comprehensive analysis and estimation system, keeping in mind the high-performance and low-cost requirements imposed on future spacecraft development.

A major problem of space-craft re-entry is the high amount of heat generated. Heat sink and ablation are the commonly used TPS, but the cost of implementing these TPSs are extremely high. The issue of heat dissipation has, therefore, been an area of extensive research. In the research article [13] a new approach of deploying minimum counter flow jet ejection as an active flow control method for heat dissipation is attempted. The concept relies on imparting sufficient momentum on the strong bow shock wave formed to induce oscillation but low enough to avoid instability during the flow interaction. Results obtained are highly promising, suggesting that a more cost-efficient way of developing TPS may have been obtained.

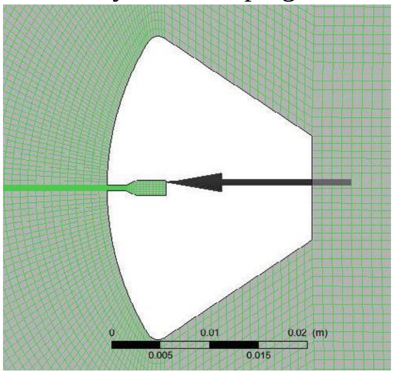


Figure 15: Mesh details with counterflow jet [13]

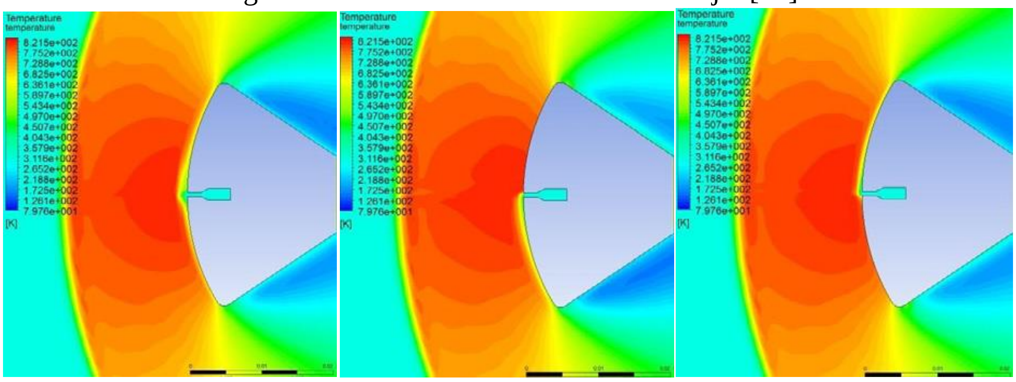


Figure 16 (a) – (c): Transient flow results of counterflow ejection at jet input velocity 10 m s^{-1} [13]

Material which is efficient to counter the heating undergone by the spacecraft during atmospheric re-entry is another problem for which attempts are being made to reach a long-term solution. Scientists in Japan have worked on the material most suitable for the thermal protection system and have proposed a new lightweight thermal protection system to reduce the re-entry weight of the capsule through the publication [14]. The article proposes a TPS which employs an ablator as the thermal protection material (TPM), and a high-temperature polyimide carbon-fibre reinforced plastic (CFRP) sandwich panel as the structure member. The proposed system led to a more than 40% weight reduction of the spacecraft, maintaining high thermal insulation and recession resistance. As per the paper [15], after conducting hypervelocity impact tests on the porous ceramic thermal tiles used in the Space Shuttle, a model using the material equation of state and strength properties of the tiles was developed that explains these findings and facilitates extrapolation to alternative tile configurations and impact conditions. The research article [16] studies how the highly porous carbon-fibre performs well against heat convection and conduction as well as possesses the property of high radiation absorption. As far as new compositions is concerned, the article [17] proposes a hybrid thermal protection system based on cork and ceramic matrix composites (CMC). These joints were obtained by high temperature inorganic adhesives and in-situ polymerization of the cork on top of CMC. This turned out to be an efficient TPS as well as possesses high values for ultimate shear strength.

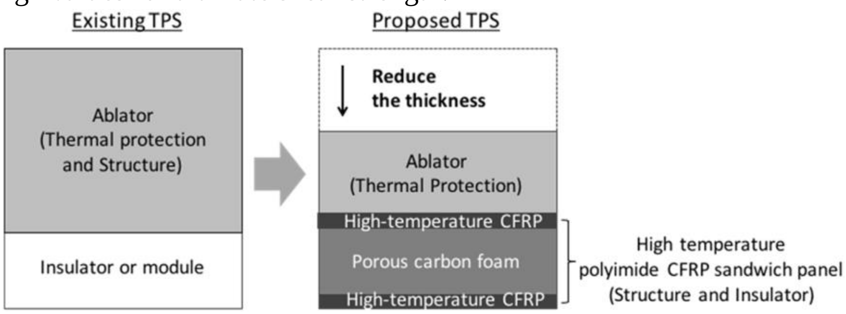


Figure 17: Concept of the proposed TPS using a high-temperature polyimide CFRP sandwich panel [14]

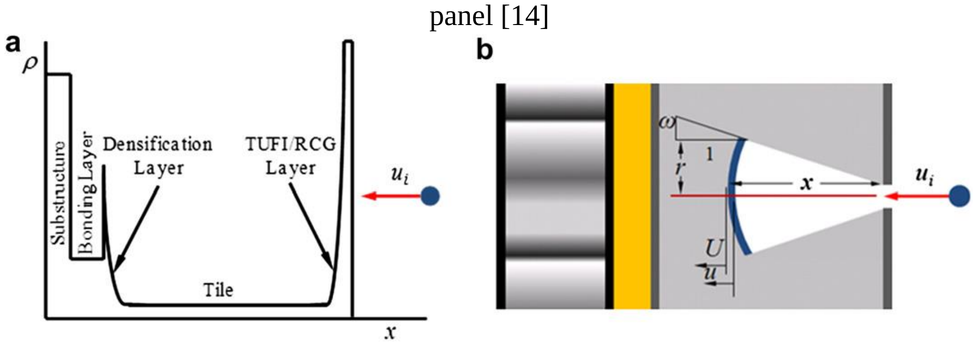


Figure 18: (a) Density profile experienced by a projectile and (b) dispersion of impact debris in the low-density tile [16]

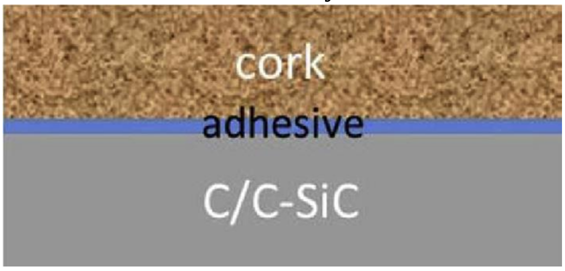


Figure 19: Schematic configuration (section view) of the joints fabricated for mechanical tests on proposed hybrid TPS made of cork and CMC [17]

5. WAY FORWARD AND INFERENCE:

During the last 50 years the access to the space changed our lives in so many ways that it is difficult to fully understand the nature of this change: if we consider technological returns – new materials, patents; in terms of economics, the space sector is an extremely profitable sector. Nano and micro-satellites also represent a huge opportunity for civilian and other activities, which has been made feasible because of micro-electronics and new materials.

However, the development and growth of the space activities and economy are closely related to the space access i.e. to TPS materials. There is no single part of the lifetime of a space system – from the launch, to the orbiting, to the re-entry – which could be enabled in absence of a thermal protection solution. Polymer Ablatives (PAs) represented and still remain the most versatile class of TPS materials. The widespread availability of some technologies related to TPS materials also constituted a breeding ground for the appearance of many small companies which are proposing new, cheap, small launchers able to offer a superior responsiveness as compared to traditional vehicles, thus to address the market request for the launch nano- and micro-satellites. These private players have also been exerting an essential role in terms of contribution to the science and technology of TPS materials. The use of nanosized fillers can also enable new approaches to exploit the properties of many materials at the base of TPS formulations. However, as a final remark on TPS materials, even if the price of nanomaterials is quickly decreasing, a systematic study must be done in the identification of the best processing techniques suitable for nanomodified TPS materials to improve their reliability and decrease the manufacturing costs [18].

REFERENCES

[1] B. R. Hollis and S. Borrelli, “Aerothermodynamics of Blunt Body Entry Vehicles,” *Progress in Aerospace Sciences,* no. 48-49, pp. 42-56, 2012.

[2] A. Riccio, F. Raimondo, A. Sellitto, V. Carandente, R. Scigliano and D. Tescione, “Optimum design of ablative thermal protection systems for atmospheric entry vehicles,” *Applied Thermal Engineering,* no. 119, pp. 541-552, 2017.

[3] J. Meseguer, I. Pérez-Grande and A. Sanz-Andrés, “Chapter 17 - Thermal Protection System,” in *Spacecraft Thermal Control*, Woodhead Publishing, 2012, pp. 305-325.

[4] A. M. Cruise, J. A. Bowles, T. J. Patrick and C. V. Goodall, “Chapter 3 - Thermal Design,” in *Principles of Space Instrument Design*, Cambridge University Press, 1998, pp. 73-156.

[5] P. Fortescue, G. Swinerd and J. Stark, “Section 11.4 - THERMAL ANALYSIS,” in *Spacecraft Systems Engineering*, John Wiley & Sons Ltd, 2011, pp. 366-371.

[6] P. Sforza, “Chapter 9 - Thermal Protection Systems,” in *Manned Spacecraft Design Principles*, Butterworth-Heinemann, 2016, pp. 391-452.

[7] A. V. Nenarokomov, O. M. Alifanov, S. A. Budnik and A. V. Netelev, “Research and development of heat flux sensor for ablative thermal protetcion of spacecrafts,” *International Journal of Heat and Mass Transfer,* no. 97, pp. 990-1000, 2016.

[8] A. Mazzaracchio and M. Marchetti, “A probabilistic sizing tool and Monte Carlo analysis for entry vehicle ablative thermal protection system,” *Acta Astronautica,* no. 66, pp. 821-835, 2010.

[9] O. M. Alifanov, V. I. Outchvatov and K. M. Pichkhadze, “Thermal Protection of Re-Entry Vehicles with the usage of Inflatable Systems,” *Acta Astronautica,* no. 53, pp. 541-546, 2003.

[10] S. R. Kannan and S. S. Katte, “Numerical investigation and correlations for heat diffusion through planar ablative thermal protection systems,” *Thermal Science and Engineering Process,* no. 7, pp. 279-287, 2018.

[11] R. T. Swann and C. M. Pittman, “Numerical Analysis of the Transient Response of Advanced Thermal Protection Systems for Atmospheric Re-Entry,” National Aeronautics and Space Administration (NASA), Washington DC, 1962.

[12] M. Guoliang and J. Guiqing, “Comprehensive analysis and estimation system on thermal environment, heat protection and thermal structure of spacecraft,” *Acta Astronautica,* no. 54, pp. 347-356, 2004.

[13] Y. Zheng, N. A. Ahmed and W. Zhang, “Heat dissipation using minimum counter flow jet ejection during spacecraft re-entry,” *Procedia Engineering,* no. 49, pp. 271-279, 2012.

[14] Y. Kobota, O. Miyamoto, T. Aoki, Y. Ishida, T. Ogasawara and S. Umezu, “New thermal protection system using high-temperature carbon fibre-reinforced plastic sandwich panel,” *Acta Astronautica,* no. 160, pp. 519-526, 2019.

[15] J. E. Miller, W. E. Bohl, E. L. Christiansen and B. A. Davis, “Ballistic performance of porous-ceramic, thermal protection systems,” *International Journal of Impact Engineering,* no. 56, pp. 40-46, 2013.

[16] A. V. Gusarov, E. Poloni, V. Shklover, A. Sologubenko, J. Leuthold, S. White and J. Lawson, “Radiative transfer in porous carbon-fiber materials for thermal protection systems,” *International Journal of Heat and Mass Transfer,* no. 144, pp. 118582-118592, 2019.

[17] K. Triantou, B. Perez, A. Marinou, S. Florez, K. Mergia, G. Vekinis, J. Barcena, W. Rotarmel, C. Zuber and A. de Montbrun, “Performance of cork and ceramic matrix composite joints for re-entry thermal protection structures,” *Composites Part B,* no. 108, pp. 270-278, 2017.

[18] M. Natali, J. M. Kenny and L. Torre, “Science and technology of polymeric ablative materials for thermal protection systems and propulsion devices: A review,” *Progress in Materials Science,* no. 84, pp. 192-275, 2016.

[19] Massachusetts Institute of Technology, *Lec 12 | MIT 16.885J Aircraft Systems Engineering, Fall 2005,* Cambridge, MA: MIT OpenCourseWare, 2008.

[20] NASA Langley Research Center, *NASA Talk - Mars Entry, Descent and Landing with Humans,* Hampton, Virgina: NASA, 2015.

[21] NASA Aeronautics Research Mission Directorate, *Technical Seminar: "Entry Descent and Landing",* Washington DC: NASA, 2007.

[22] NASA Aeronautics Research Mission Directorate, *Technical Seminar: "Thermal Protection Systems",* Washington DC: NASA, 2007.

APPENDIX A – IMPORTANT FIGURES

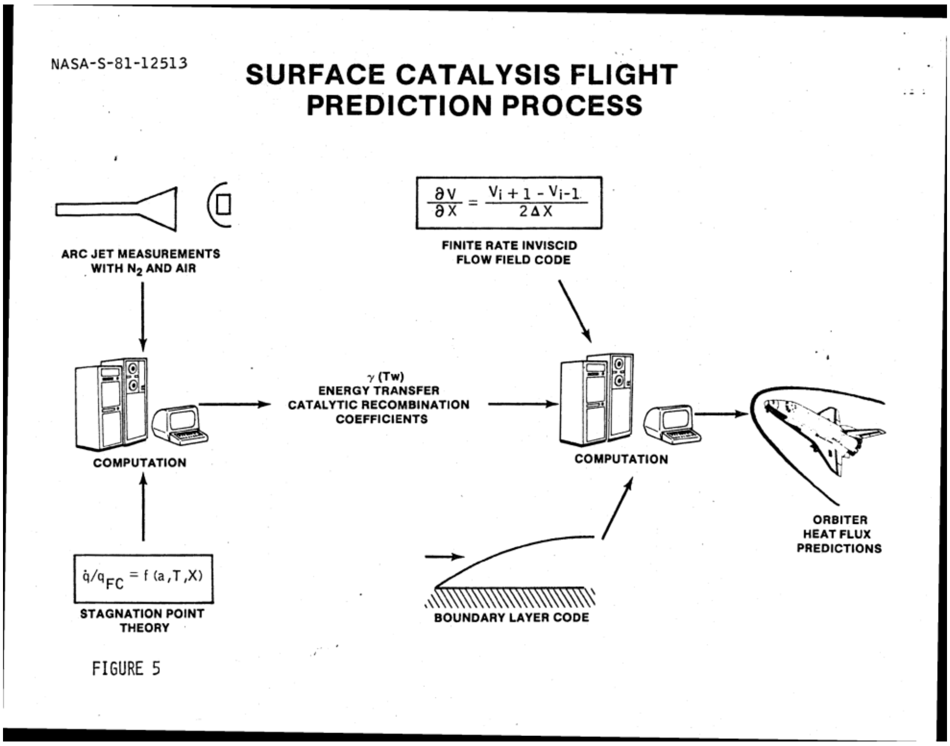


Figure 20: Figure detailing process to predict a spacecraft's (here, orbiter) heat flux predictions [19]

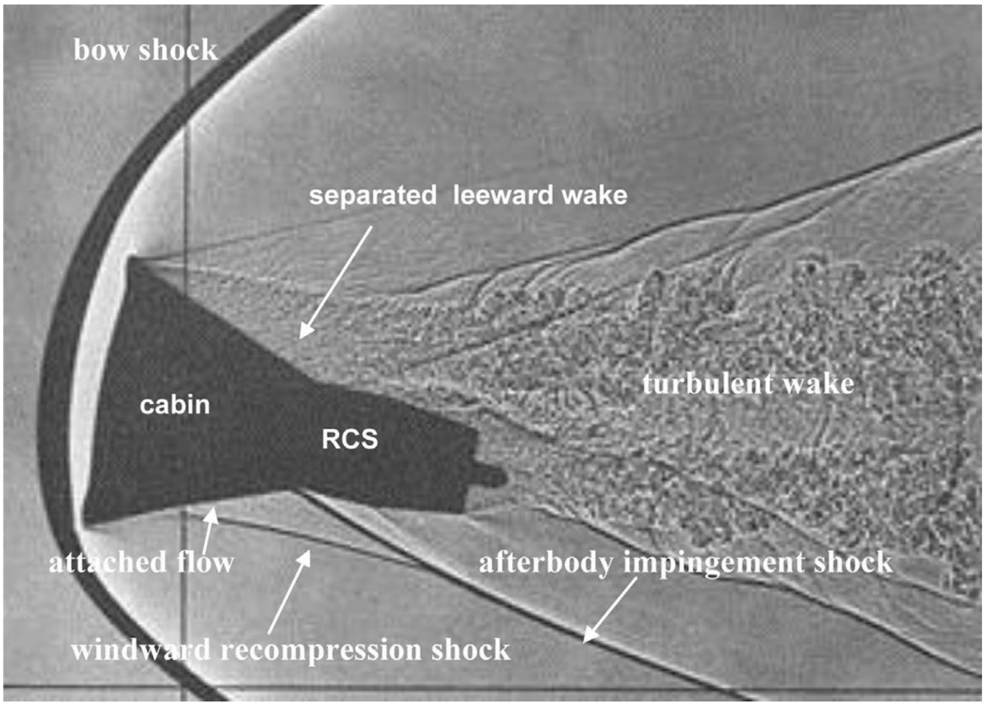


Figure 21: A Gemini capsule model undergoing a transient flow experiment to test thermal phenomenon (NASA)

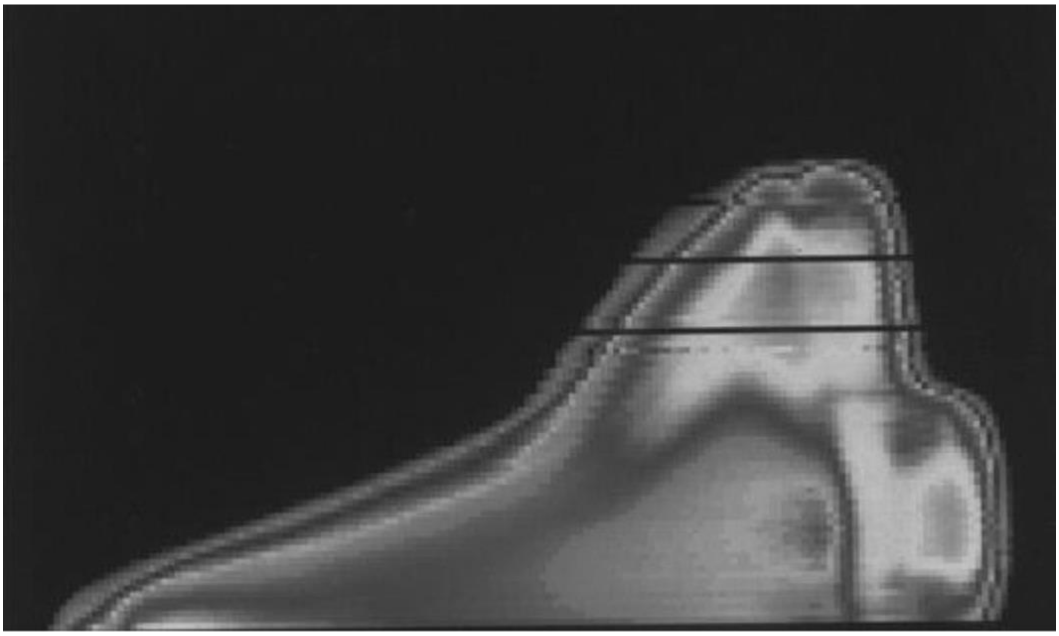


Figure 22: Distribution of heat flux on the Space Shuttle Orbiter during re-entry (NASA)

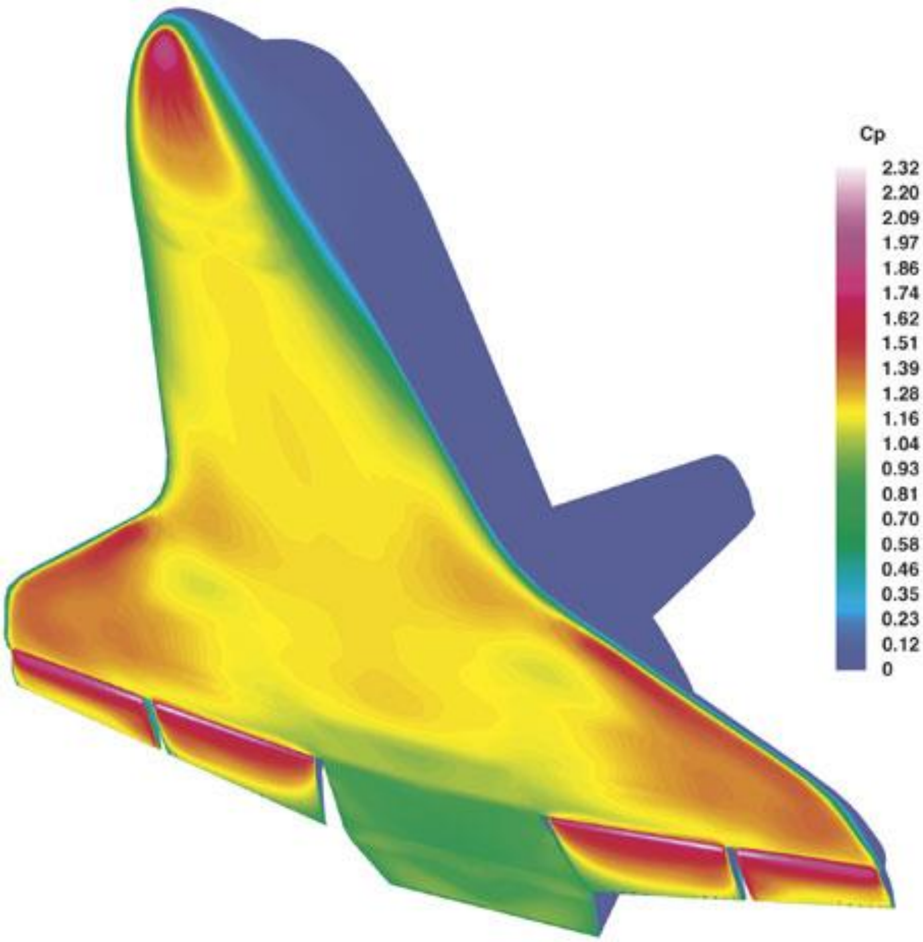


Figure 23: Simulation of heat flux distribution of the Space Shuttle on computer software (NASA)

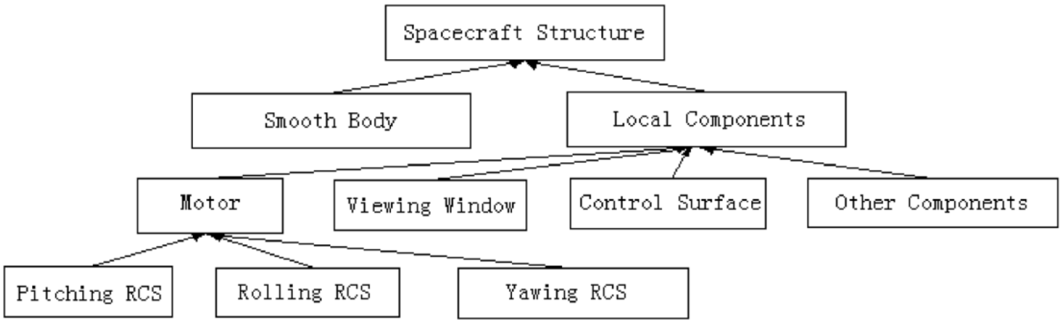


Figure 24: Structure of aerothermodynamics database and interrelation [12]

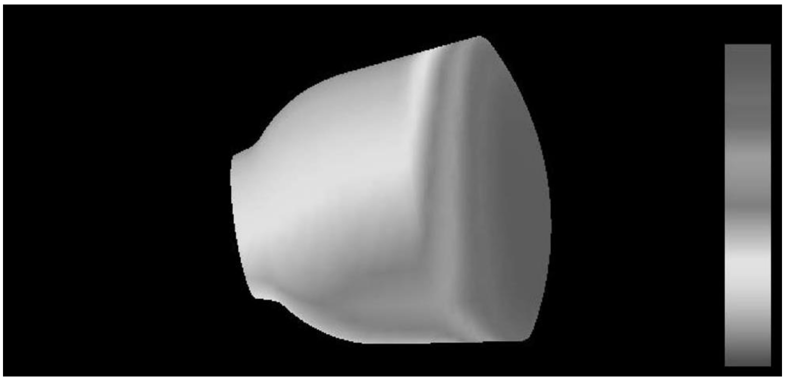


Figure 25: Surface temperature contours of smooth body [12]

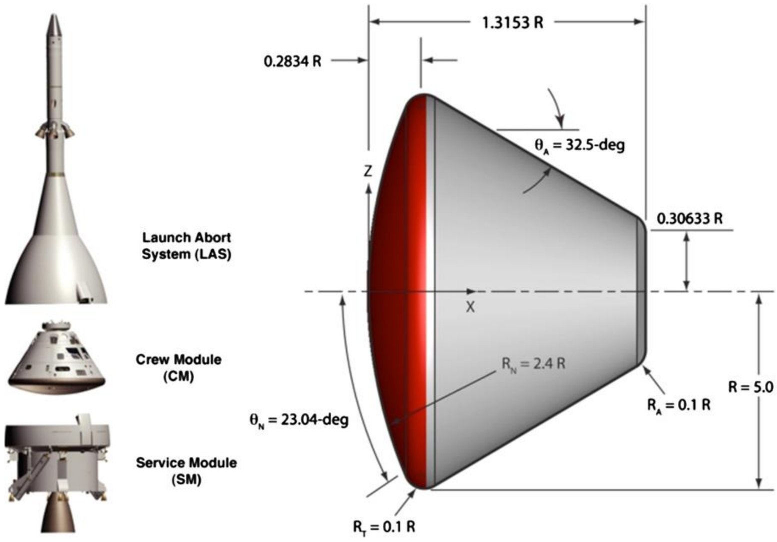


Figure 26: Orion CEV crew module dimensions. We must note that the heat shield of the crew module is highlighted in red. [1]

APPENDIX B – IMPORTANT TABLES AND GRAPHS

Table 1
The specific heat capacities of some common substances (in J/kg) [6]

Aluminium alloy	920
Copper	420
Stainless steel	500
Titanium	523
Concrete	3350
PTFE	1050
Ceramic	800
Nylon	1680
Carbon fibre	840
Water	4190
Air @ STP	993
Helium	5250

Graph 1: Predicted and Experimental Data of Heat Flux encountered by the Space Shuttle Orbiter during STS-3 Mission (MIT, USA) [19]

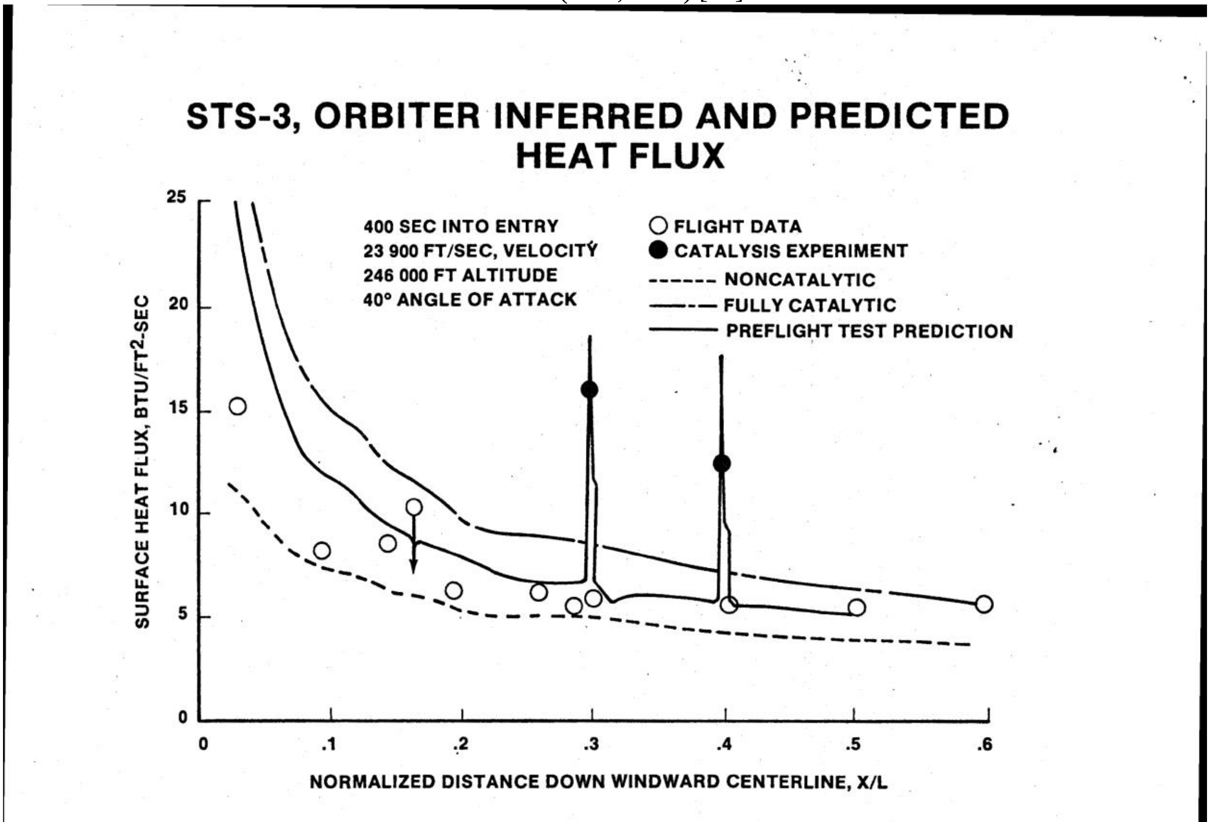
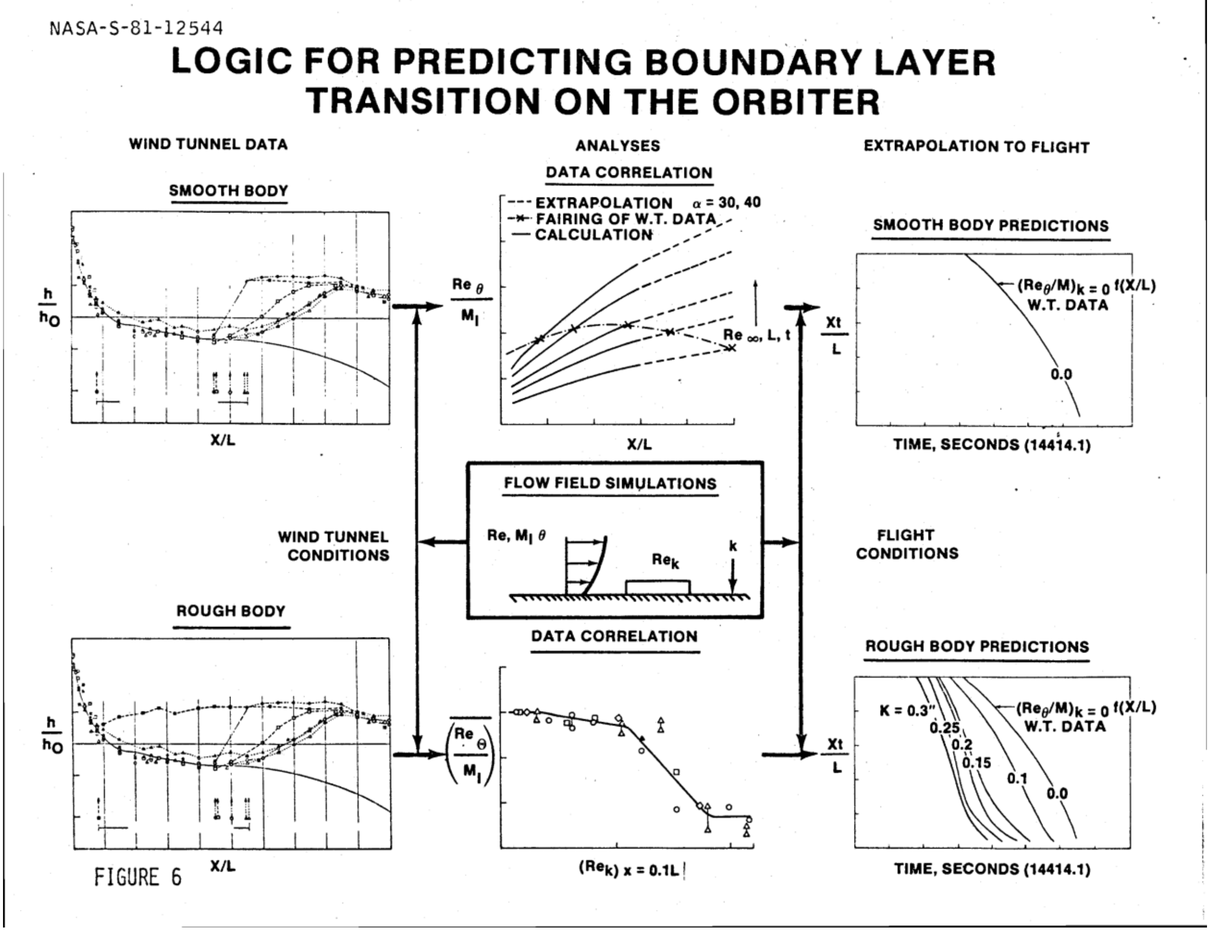
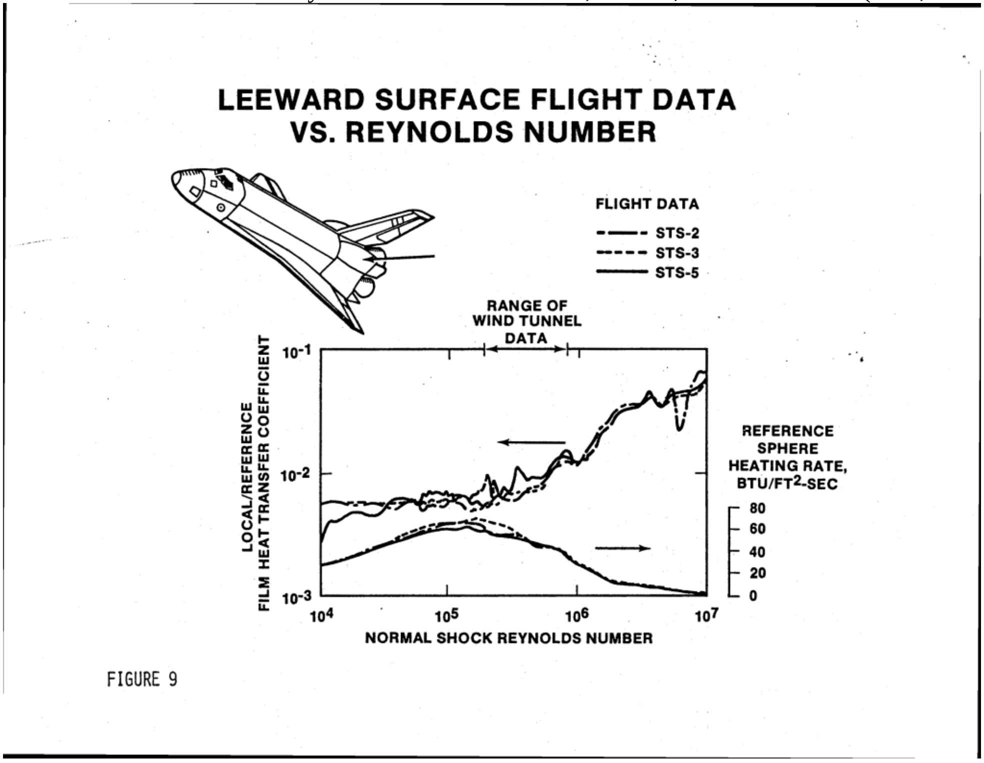


FIGURE 8

Graph 2: Data Required to Predict Boundary Layer Transition on Orbiter (MIT, USA) [19]



Graph 3: Surface Heat Data vs. Reynolds Number for STS-2, STS -3, STS-5 missions (MIT, USA) [19]



Graph 4, 5, 6, 7: CEV Mach 10, α (Angle of attack) = 28.deg data and comparison with AEDC Tunnel 9 test [1]

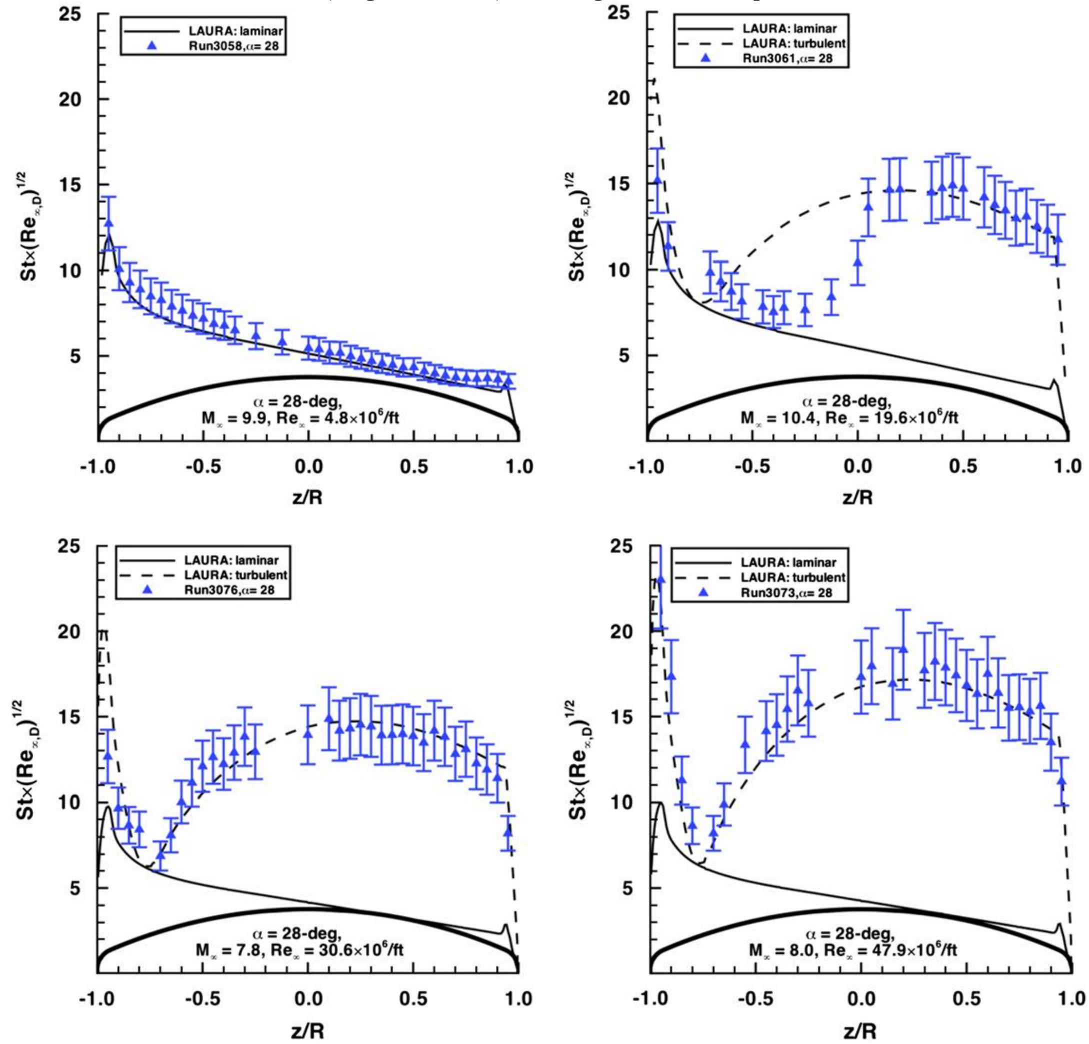
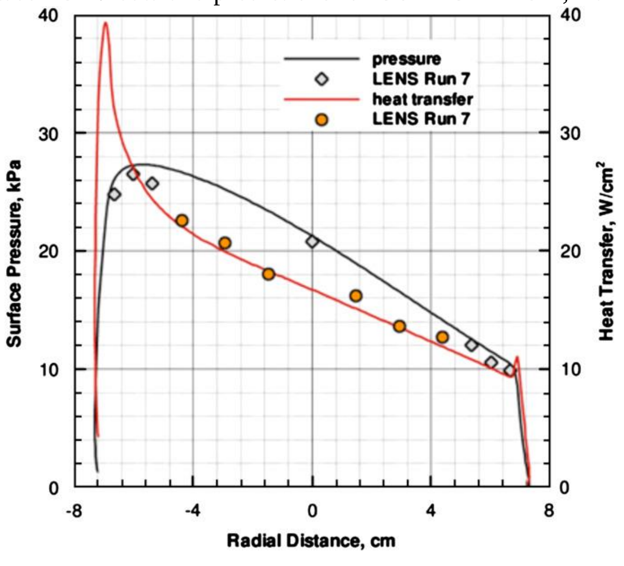


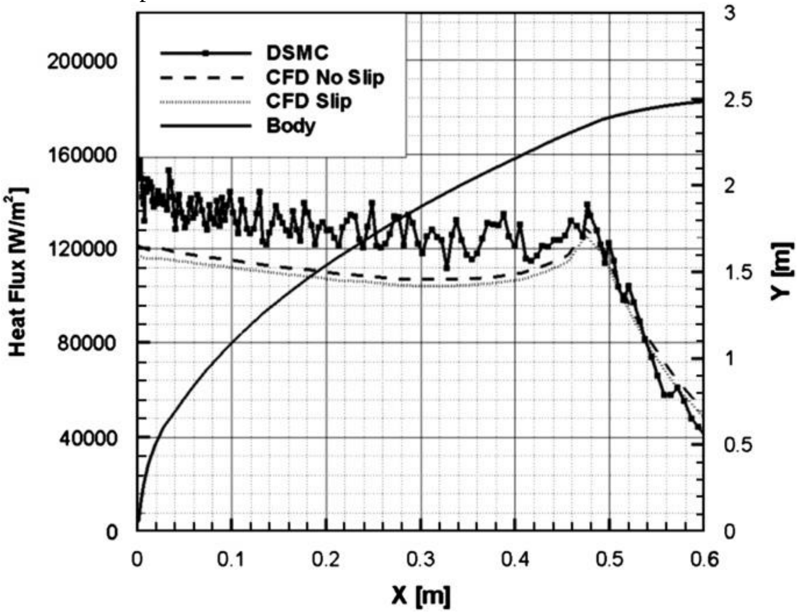
Table 2: Orion CEV high-enthalpy test conditions for CUBRC LENS-1 [1]

Run	α (deg.)	T_∞, $T_{V\infty}$ (K)	U_∞ (m/s)	H_0 (MJ/kg)	$\rho_{\infty,N_2}Z$ (kg/m^3)	$\rho_{\infty,O_2}Z$ (kg/m^3)	$\rho_{\infty,\,NO}$ (kg/m^3)	$\rho_{\infty,\,O}$ (kg/m^3)
7	28	57	1805	1.7	6.798E−03	2.144E−03	0.000E+00	0.000E+00
8	28	191	2949	4.8	2.431E−03	6.529E−04	2.339E−04	1.902E−06
9	28	494	4054	9.2	1.214E−03	3.035E−04	9.108E−05	3.402E−05
10	28	631	4601	12.4	7.939E−04	1.493E−04	6.347E−05	6.949E−05

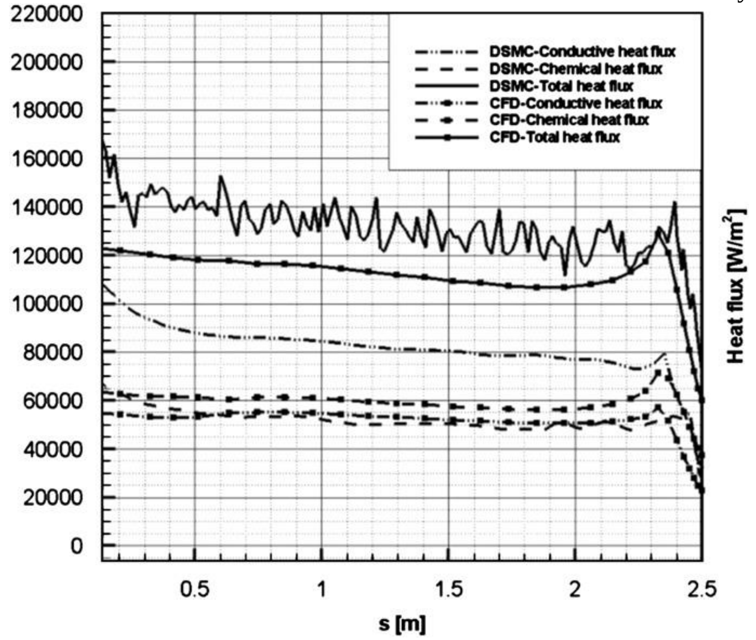
Graph 9: Comparison between CEV data and predictions for CUBRC LENS-1, Run 7, 1.7 MJ/kg enthalpy [1]



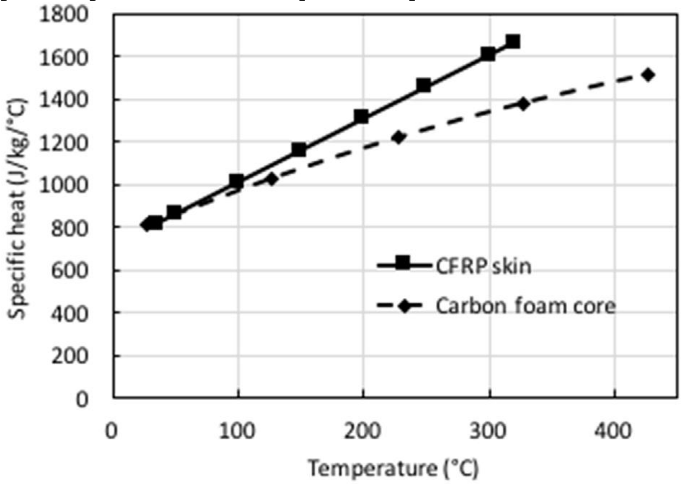
Graph 10: Predicted CEV heat flux at 85 km altitude [1]



Graph 11: Chemical and conductive contributions to CEV heat flux at 85 km with fully catalytic wall [1]



Graph 12: Specific heat of skin panel and porous carbon foam core [14]



APPENDIX C – IMPORTANT EQUATIONS

EQUATION SET 1 [7]:

Nomenclature

Latin symbols

A	Arrenious coefficient
b	length of heat flux sensor
$c(T)$	heat capacity
$C_g(T)$	heat capacity of gas
E	energy of activation
$f_m(\tau)$	temperature measurements
$H(T)$	heat effect
$\overline{g^s}$	vector of gradient minimization method at current iteration
J	leas-square minimized functional
$\overline{J_p'^{(s)}}$	gradient of the functional J at current iteration
M	number of thermocouples
n	power of thermo-kinetic reaction
$\bar{p}$	vector of unknown parameters
$q_1(\tau)$	heat flux at the heated boundary
$q_2(\tau)$	heat flux at the inner boundary
$q_{w_i}(\tau)$	heat flux at selected points at the surface
Q_{electr}	electrical heat at the heater
$T(\tau, x)$	temperature
$T_0(x)$	initial temperature
$T_{w_i}(\tau)$	temperature at selected points at the surface
x	spatial variable
$x_1, x_2, \ldots, x_m$	coordinate of thermocouples

Greek symbols

$\alpha_1, \alpha_2, \beta_1, \beta_2$	coefficients determined kind of boundary conditions (1 for left, 2 for right)
$\overline{\gamma^s}$	step descent
$\Delta T(x, \tau)$	increment of temperature
δ_f	integral error of temperature measurements
$\Phi(x, \tau)$	adjoint variable for density
$\lambda(T)$	thermal conductivity
$\rho(x, \tau)$	current density of material
ρ_0	initial density of virgin material
ρ_c	density after thermokinetic process
$\sigma_m(\tau)$	measurement variance
τ	time
τ_m	final time
τ_r	time of the beginning of thermokinetic process
τ_c	time of the end of thermokinetic process
$\psi(x, \tau)$	adjoint variable for temperature
$\theta(x, \tau)$	increment of density

Inverse Problem Algorithm:

$$C\big(T(\tau,x)\big)\rho\frac{\partial T(\tau,x)}{\partial\tau} = \frac{\partial}{\partial x}\left(\lambda\big(T(\tau,x)\big)\frac{\partial T(\tau,x)}{\partial\tau}\right) + C_g\big(T(\tau,x)\big)\int_x^L \frac{\partial\rho(\xi,\tau)}{\partial\tau}d\xi\,\frac{\partial T(\tau,x)}{\partial x} + H\big(T(\tau,x)\big)\frac{\partial\rho(x,\tau)}{\partial\tau} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (22)$$

$$T(0,x) = T_0(x), \quad 0 < x < L \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (23)$$

$$-\alpha_1\lambda\big(T(0,\tau)\big)\frac{\partial T(0,\tau)}{\partial x} + \beta_1 T(0,\tau) = q_1(T(0,\tau),\tau) \ldots\ldots\ldots\ldots\ldots\ldots\ldots (24)$$

$$-\alpha_2\lambda\big(T(L,\tau)\big)\frac{\partial T(L,\tau)}{\partial x} + \beta_2 T(L,\tau) = q_2(T(L,\tau),\tau) \ldots\ldots\ldots\ldots\ldots\ldots\ldots (25)$$

$$\rho(x,\tau_r) = \rho_0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (26)$$

$$\rho(x,\tau_c) = \rho_c \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (27)$$

$$\frac{\partial\rho(x,\tau)}{\partial\tau} = \left\{\begin{matrix} 0, & T(x,\tau) < T_r & \\ -\rho^n A e^{-\frac{E}{RT(x,\tau)}}, & \rho(x,\tau) > \rho_c, & T(x,\tau) \geq T_r \\ 0, & \rho(x,\tau) \leq \rho_c & \end{matrix}\right\} \ldots\ldots (28)$$

$$T^{\exp}\quad(x_m,\tau) = f_m(\tau) \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (29)$$

$$J\big(q_1(\tau)\big) = J(\bar{p}) = \sum_{m=1}^{M}\int_{\tau_{min}}^{\tau_{max}}\big(T(x_m,\tau) - f_m(\tau)\big)^2 d\tau \ldots\ldots\ldots\ldots\ldots\ldots\ldots (30)$$

To construct an iterative algorithm of the inverse problem solving a conjugate gradient method is used. The successive approximations process is constructed as follows:

1. a priori, an initial approximation of the unknown parameter vector is set.
2. a value of the unknown vector at the next iteration are calculated.

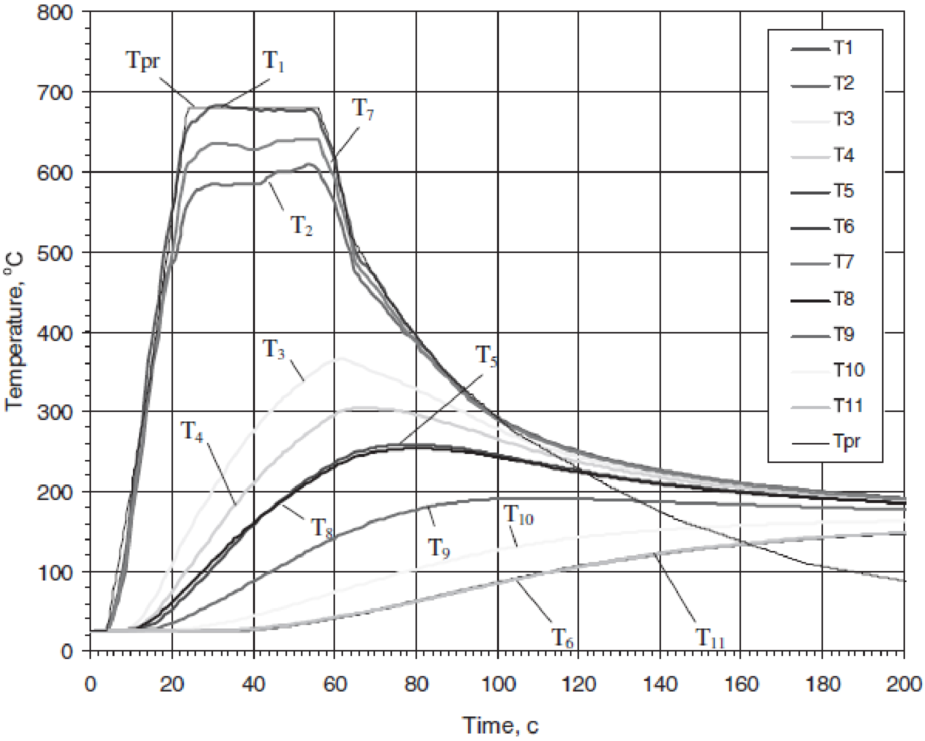
The greatest difficulties in realizing the gradient methods are connected with calculation of the minimized functional gradient. In the approach being developed the methods of calculus of variations are used. Here an analytic expression for the minimized function gradient has been obtained:

$$J'_{q_1} = \psi_1(x_0,\tau) \ldots\ldots\ldots\ldots\ldots (31)$$

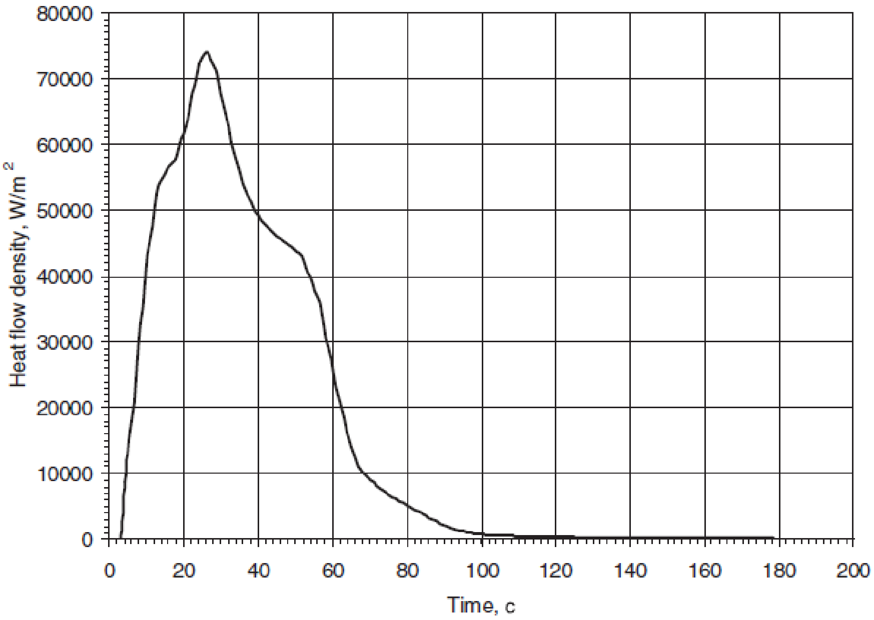
where $\psi(x,\tau)$ – solution of a boundary-value problem adjoint to a linearized form of the direct problem (1) – (8).

$$\frac{\partial \psi_m}{\partial \tau}\rho_m C + \frac{\partial}{\partial x}\left(\lambda \frac{\partial \psi_m}{\partial x}\right) - \left(\frac{d\lambda}{dT}\frac{\partial T_m}{\partial x} + C_g \int_x^L \frac{\partial \rho_m}{\partial \tau} d\xi\right)\frac{\partial \psi_m}{\partial x} + \left(\frac{dH}{dT}\frac{\partial \rho_m}{\partial \tau} + \frac{\partial \rho_m}{\partial \tau}C - C_g\frac{\partial \rho_m}{\partial \tau}\right)\psi_m + \phi_m \frac{\partial F}{\partial T} = 0 \dots\dots\dots (32)$$

Results obtained:



Graph 13: The results of temperature measurements into D_{2A} and D_{2B} sensors



Graph 14: The density of thermal flux on TPS

EQUATION SET 2 [8]:

Nomenclature

A	see Eq. (10)
A_i, B_i, C_i, D_i	coefficients; see Eq. (16)
a, b	exponents, dependent on ρ_{atm} and V
B	activation temperature, K
C_c	constant, dependent on the atmosphere
C_r	constant, dependent on the atmosphere
c_p	specific heat, J/(kg K)
dt	integration time step, s
F	exterior view factor
$f(V)$	function, dependent on the velocity
H_d	pyrolysis enthalpy, J/kg
h	enthalpy, J/kg
$\overline{h}$	partial heat of charring, J/kg
h_w	wall enthalpy, J/kg
h_0	total enthalpy, J/kg
i	node index [1–N]
J	material index
j	sample index [1–M]
K	collision frequency factor, kg/(m^3 s)
k	thermal conductivity, W/(m K)
M	number of samples
$\dot{m}_c$	char removal rate, kg/(m^2 s)
$\dot{m}_g$	pyrolysis gas mass flow rate, kg/(m^2 s)
N	number of nodes
n	decomposition reaction order
$\dot{Q}_{in}$	net total heat flux at surface, W/m^2
$\dot{q}_{c,blow}$	net hot wall convective heat flux, W/m^2
$\dot{q}_{comb}$	combustion heat flux, W/m^2
$\dot{q}_{c,w}$	cold wall convective heat flux, W/m^2
$\dot{q}_R$	internal radiative heat flux, W/m^2
$\dot{q}_{rad}$	radiative heat flux, W/m^2
r_n	vehicle nose radius, m
S	surface recession, m
$\dot{S}$	char recession rate, m/s
T	temperature, K
T_{abl}	ablation (start) temperature, K
T_{char}	charring temperature, K
T_w	wall temperature, K
T_∞	freestream temperature, K
t	time, s
V	velocity, m/s^2
x	mobile coordinate system, y–S, m
y	fixed coordinate system, m
ΔH_c	heat of combustion per unit weight, J/kg
ΔX	node thickness, m
α_c	weighting factor for char mass loss
α_g	weighting factor for pyrolysis gases
ε_w	surface emissivity
ρ	density, kg/m^3
ρ_{atm}	atmospheric density, kg/m^3
σ	standard deviation; or Stefan–Boltzmann constant in Eq. (9), W/(m^2 K^4)
τ	mass fraction of virgin material

Subscripts

BL	bond-line
c	char
g	pyrolysis gas
v	virgin material

Superscripts

$\overline{}$	mean value
$'$	value at time $t+dt$

Model Assumptions

1. The ablative material decomposes from the virgin state to the porous char in the reaction zone. The pyrolysis gases evolve exclusively from this region.
2. The reaction zone is defined by an inferior temperature T_{abl} and a superior temperature T_{char}. These limit temperatures are determined for every type of material by evaluating available experimental thermo-gravimetric curves.
3. The gases that are generated in the reaction zone pass through the upper layers without a pressure drop and are injected in the upper boundary layer.
4. A local thermal equilibrium exists between the gases and the porous char
5. The generated pyrolysis gases do not react with other materials
6. There is no thermal hysteresis in the char's properties, particularly the conductivity
7. The interface with the inner cabin environment is adiabatic.

Equations involving the Monte Carlo Analysis:

The internal energy balance is expressed by the given equation,

$$\rho c_p \frac{\partial T}{\partial t}\bigg|_x = \frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x} - \dot{q}_R\right)\bigg|_t + (H_d - h)\frac{\partial \rho}{\partial t}\bigg|_y + \dot{S}\rho c_p \frac{\partial T}{\partial x}\bigg|_t + \dot{m}_g \frac{\partial H_d}{\partial x}\bigg|_t \dots\dots\dots\dots (33)$$

For both the virgin material and the char, the local specific heat depends on temperature, whereas the local thermal conductivity depends on temperature and pressure. In the reaction zone, the expressions, which use the classic 'mixing rule', are as follows:

$$c_p = \tau c_{pv} + (1-\tau)c_{pc} \dots\dots\dots\dots\dots\dots (34)$$

$$k = \tau k_v + (1-\tau)k_c \dots\dots\dots\dots\dots\dots\dots (35)$$

where τ is the mass fraction of the virgin material.

$$\tau = \frac{\left(1 - \frac{\rho_c}{\rho}\right)}{1 - \frac{\rho_c}{\rho_v}} \dots\dots\dots\dots\dots\dots (36)$$

For continuity, the following expression is assumed for the density in the partially charred reaction zone,

$$\rho = (\rho_v - \rho_c)\left(\frac{T - T_{abl}}{T_{abl} - T_{char}}\right) \dots\dots\dots\dots\dots (37)$$

The pyrolysis gas enthalpy H_d also depends on temperature and pressure, whereas the partial heat of charring can be represented as,

$$\bar{h} = \frac{\rho_v h_v - \rho_c h_c}{\rho_v - \rho_c} \dots\dots\dots\dots\dots\dots (38)$$

A large-scale model is used to represent the pyrolysis decomposition. It adopts the Arrhenius equation with relevant coefficients for the entire composite.

$$\frac{\partial \rho}{\partial t} = K\left(\frac{\rho - \rho_c}{\rho_v - \rho_c}\right)^n \exp -B/T \dots\dots\dots\dots\dots\dots (39)$$

The internal decomposition transforms part of the solid into pyrolysis gases. Due to the hypotheses of quasi-static one-dimensional flow and non-permeability of the interface with the material zone, the pyrolysis gas mass flow is associated with decomposition by the following:

$$\frac{\partial \dot{m}_g}{\partial y} = \frac{\partial \rho}{\partial t} \dots\dots\dots\dots\dots\dots (40)$$

Referring to Equation (16) in 2.2 of the paper, the equation splits the aerodynamic heating into two components, each of which is handled differently. Such differentiation is necessary as the convective part can be significantly reduced by the injection of pyrolysis gases into the boundary layer.

The Monte Carlo Analysis provides us with data as follows:

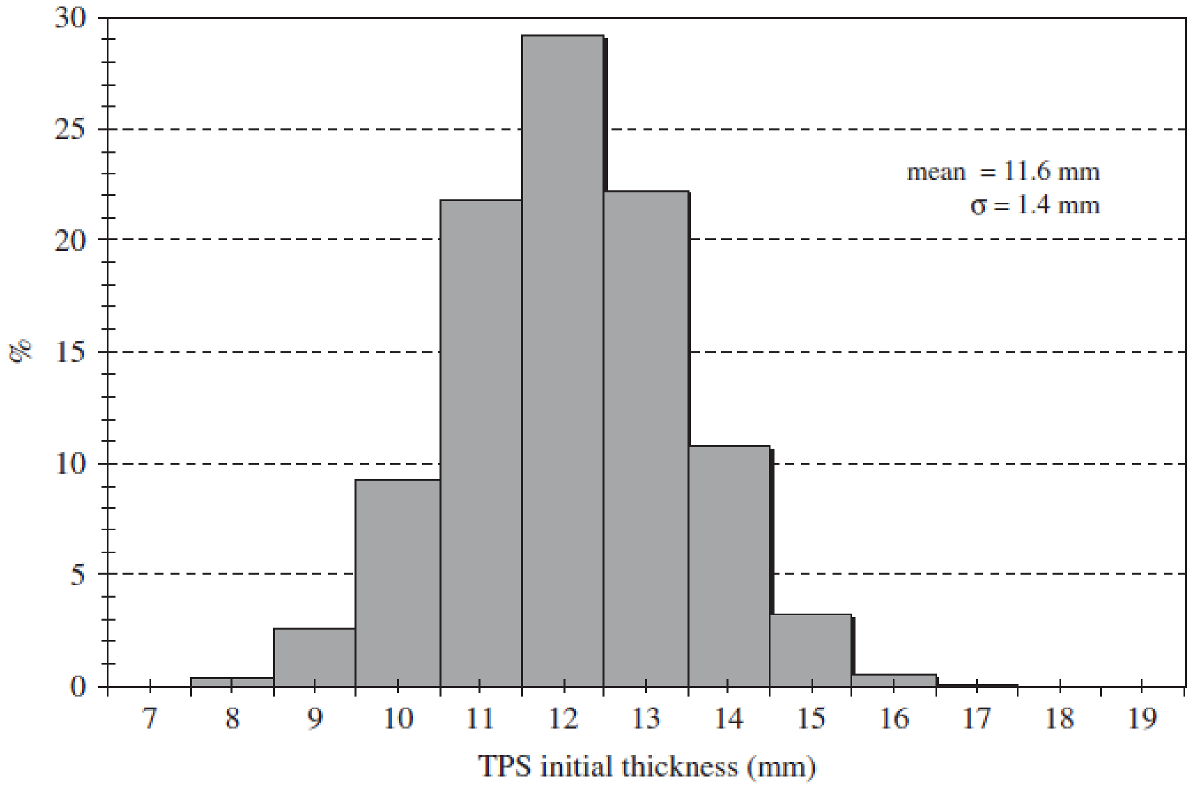


Figure 27: SLA-561V TPS thickness probability distribution for Mars Exploration Rover [8] (NASA)

APPENDIX D – THERMAL DESIGN DETERMINATION

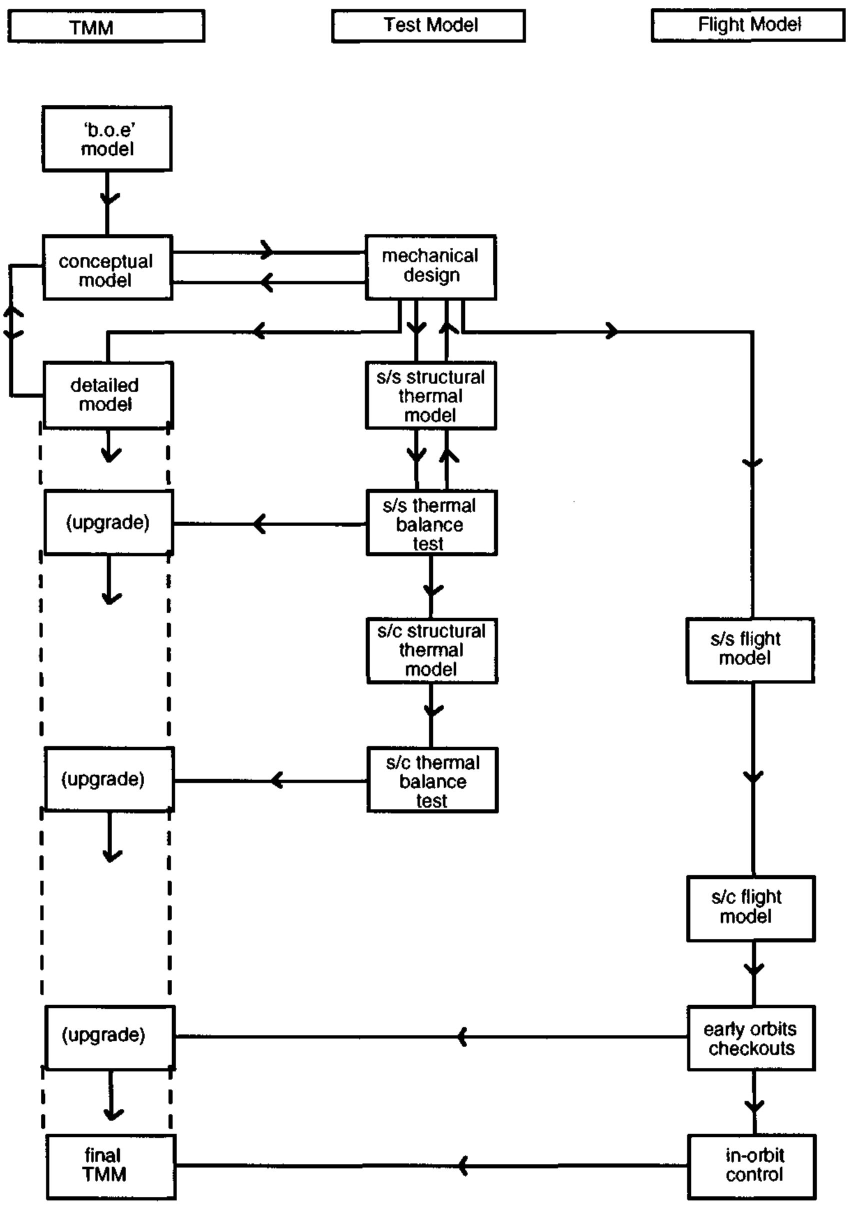


Figure 28: Strategic flow chart for thermal design determination [4]

I. *B.O.E. models* [4]

- calculate the mean temperature of the experiment given the total internal power dissipation and the total radiating area to space. This should indicate if your experiment requires overall cooling or overall heating
- if overall cooling you may need to propose a radiating surface separate from the experiment that it serves
- if overall heating is required quantify this bearing in mind that you will probably need to minimise radiation losses by selection of a low emissivity outer surface. MLI may need to be proposed
- consider what are the likely materials from which the experiment will be constructed. If possible decide if the model is likely to be predominantly conductive or predominantly radiative

- set up a small approximate model with estimated conductive and radiative links but without indirect radiative links
- determine the main heat flow paths
- determine where the main thermal gradients occur

II. *Conceptual models* [4]

- carefully consider the nodal designations
    - do not have too many nodes; the model needs to have a short run-time and to be understandable
    - ensure you have adequate numbers of nodes characterising regions of significant thermal gradients
    - give some thought as to the likely nodal map of the detailed model; it is useful to have nodal boundaries which coincide with those of other models where possible and greatly simplifies the compatibility analysis between any two models
- consider at an early stage if any subsystem of the experiment can be thermally autonomous; if so, this makes for a tidier, more secure thermal design because it minimises interdependence and simplifies subsystem testing. If subsystems can be autonomous build in the appropriate thermal isolation between system and subsystem
- if heaters are required consider the practicalities of how and where they will be located
- if heater mats are to be used and are of a size comparable with the nodes on which they are to be located, reduce the conductive links to the heater nodes appropriately since the heaters will short out part of the links. It may be necessary to reappraise the local nodal network
- perform a common sense check on the output from the model. Do not proceed with the model development until you are satisfied with your physical understanding of the model; propagated errors in a model become increasingly difficult to identify
- construct a temperature map and a heat flow map as a format for presenting, summarising and understanding the model
- for any heat flow map, check that the net heat flow to any node which is *not* a boundary is zero to within the power resolution of the map
- divide large models into sub-models wherever possible. Represent sub-models by isothermal boundaries in the larger model. This makes for a more manageable design in the early stages
- minimise the changes between successive models; ideally limit this to one change so that the effects of that single change are manifest
- keep a log of all model runs and note the changes from previous models
- have a baseline model against which development models are tested. Upgrade the baseline model when it is acceptable. Keep a record of input and output data for all the significant (i.e. baselined) models
- remember to model the external heat input for steady-state models. This can be done as an averaged heat input to the appropriate nodes
- generate a transient model, particularly if the orbit is highly eccentric
- to assist in the sizing of heaters (or coolers) set the appropriate node at the required temperature and make it an isothermal or boundary node; the model should then indicate the constraining power required to hold that node at that constant temperature
- set up boundary nodes to represent external surfaces. It is useful to have a number of these to avoid their 'shorting' any parallel nodes and are useful in exploring the sensitivity of the system to asymmetries in the external thermal environment
- perform sensitivity analyses in order to identify critical areas of the design
- use the model to identify the locations of inflight thermal sensors that representatively monitor the thermal health of the experiment

III. *Detailed model*

- attempt to make the nodal map of the detailed model compatible with that of the conceptual model
- ensure that the critical areas and the sensitive elements as identified by the conceptual model receive particular attention in the detailed model
- keep detailed notes as to all calculations of all links
- if there are any autonomous thermal subsystems, model these separately and perform consistency checks against the corresponding conceptual model
- at the earliest stage carry out compatibility tests between the full detailed model and the full conceptual model. Update both as necessary since it is useful to carry forward a representative conceptual model even after the detailed model has been constructed
- use temperature maps and heat flow maps as an aid to understanding the characteristics of the model
- if the detailed model indicates that changes in the design are required, use the conceptual model to develop the changes and then transcribe into the detailed model

IV. Thermal balance tests

- use the detailed model to predict the thermal performance of the experiment in the particular thermal balance test facility
- if boundary temperatures have to be established as part of the hardware conductive interface, carefully consider how these temperatures are to be established in terms of the power flows required to maintain these temperatures
- ensure that boundary temperatures are accurately measured and monitored, particularly if the power flows to the boundaries are high.
- if power flows can be measured (e.g. by the use of calibrated thermal impedances), then design the test hardware to include this
- carefully consider the locations of additional thermistors to ensure a good knowledge of the thermal characteristics in regions of large temperature gradients, large heat flows and important mechanical/thermal joints and interfaces
- use the test data to constrain the boundaries of a sub part of the full model to check the model in that vicinity and to determine the likely major heat flow paths
- ensure that all the thermistors are adequately calibrated in the temperature region over which the particular thermistors are to be used
- with the system in thermal equilibrium in air at room temperature and with no power dissipations ensure that all thermistors are recording the same room temperature
- ensure that the positions and types of all thermistors are properly documented in a log and on drawings prior to the test; photographs are particularly helpful
- model any mechanical support for the experiment and any electrical loom to the chamber walls particularly if they represent significant heat flow paths
- carefully select in advance the thermal stability criteria to be used as the basis for terminating the test.
- ensure that these criteria are met or that there are good grounds for changing them;
- it is better to have steady-state temperatures from one test than non-steady-state
- temperatures from a number of tests
- remember that the thermal balance test is to test the model; do not be over concerned if the test is unrepresentative of the flight conditions provided that it will maximise the return of the test in terms of understanding the experiment through the model.
- notwithstanding the above make one of the thermal balance test runs as fully representative of flight to obtain an experimental verification of the baseline thermal characteristics of the experiment independently of the model
- if at all possible ensure that there is sufficient time in the thermal balance test to get results from more than one power set-up

- keep real-time time plots of the key temperatures; these can be used to judge if the predicted power inputs are adequate and for extrapolation purposes to predict when the test termination criteria are likely to be met
- remember that a successful outcome from a thermal balance test is essential; if the implications of the results are inconclusive do not baulk at the prospect of a re-test